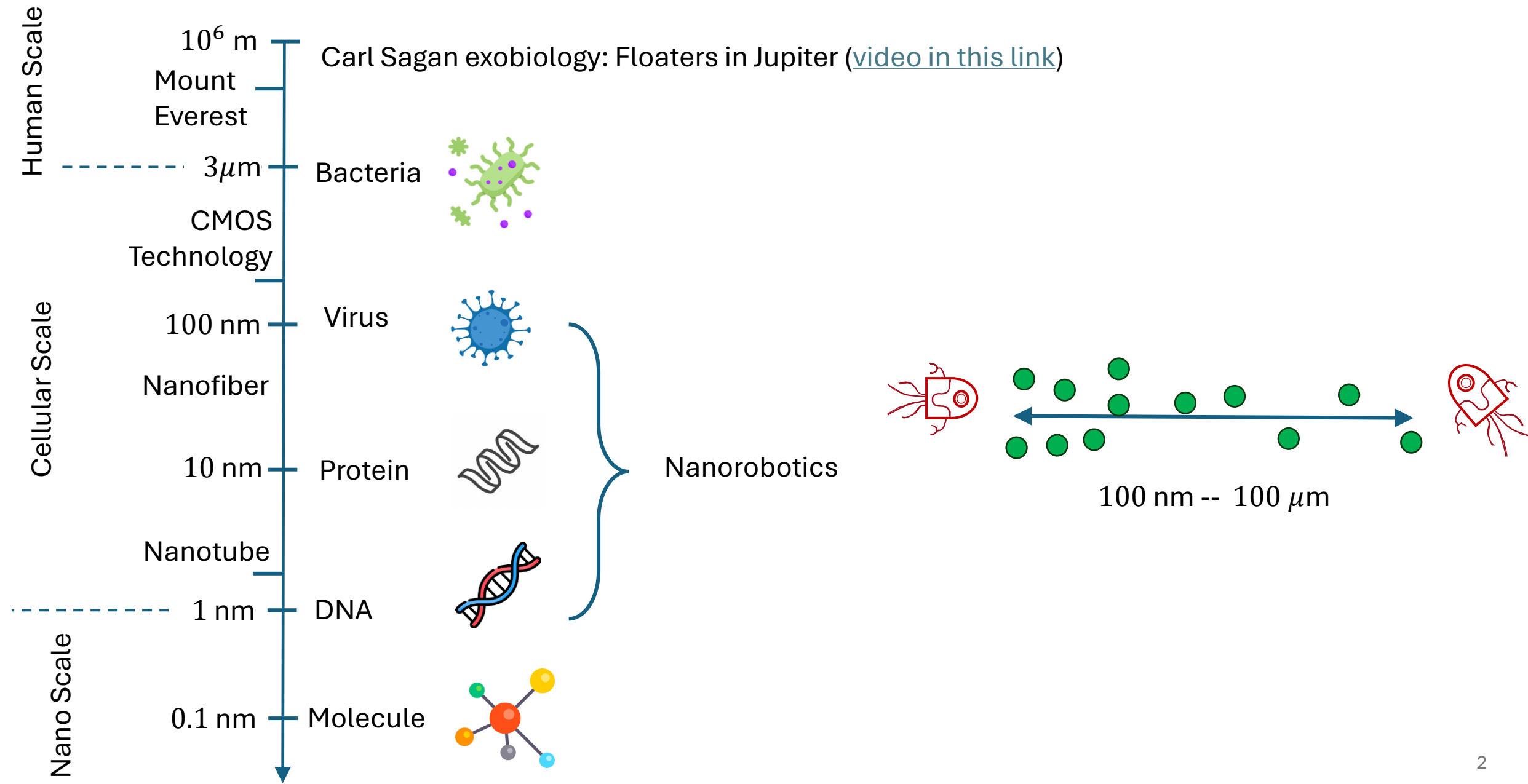
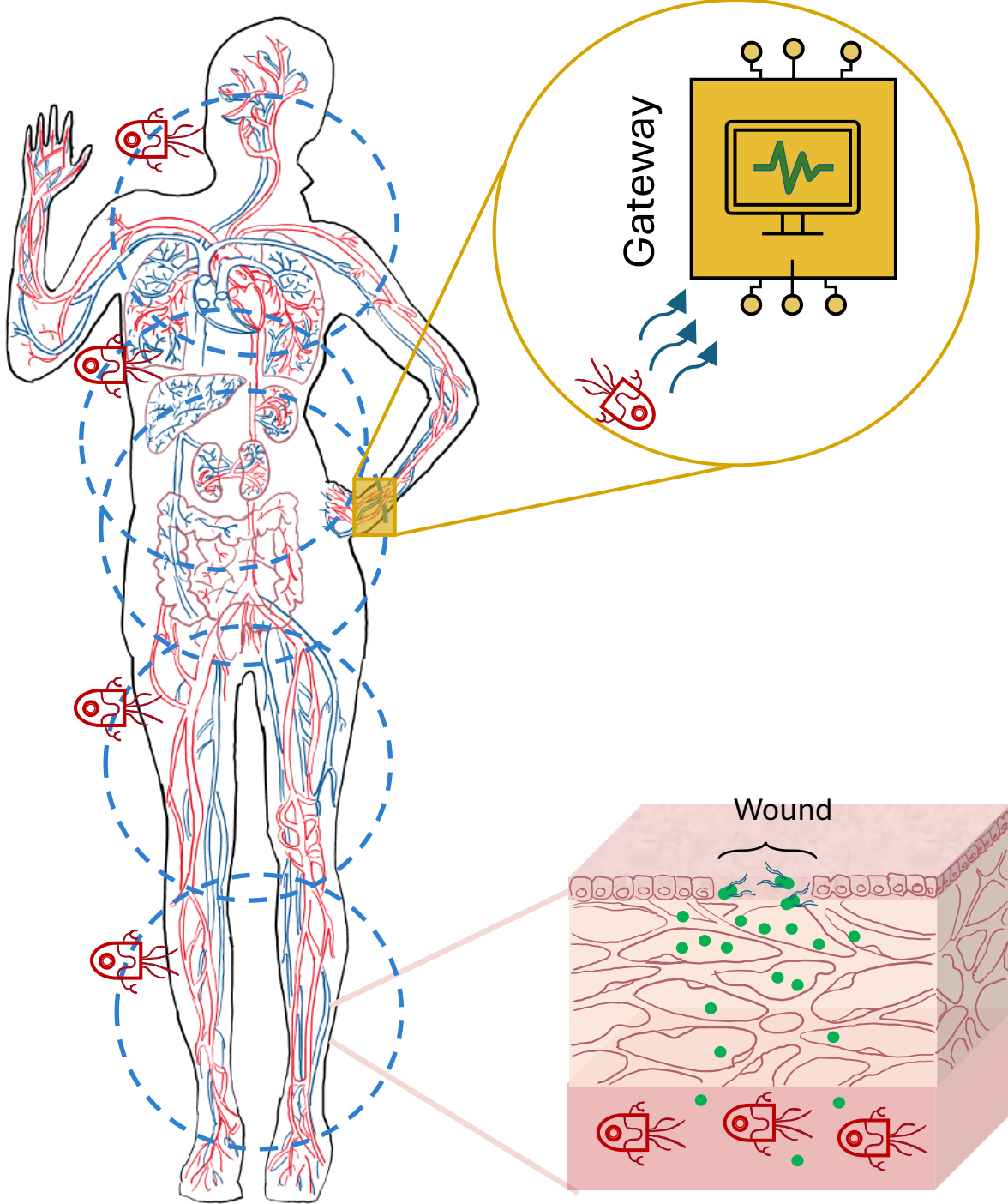


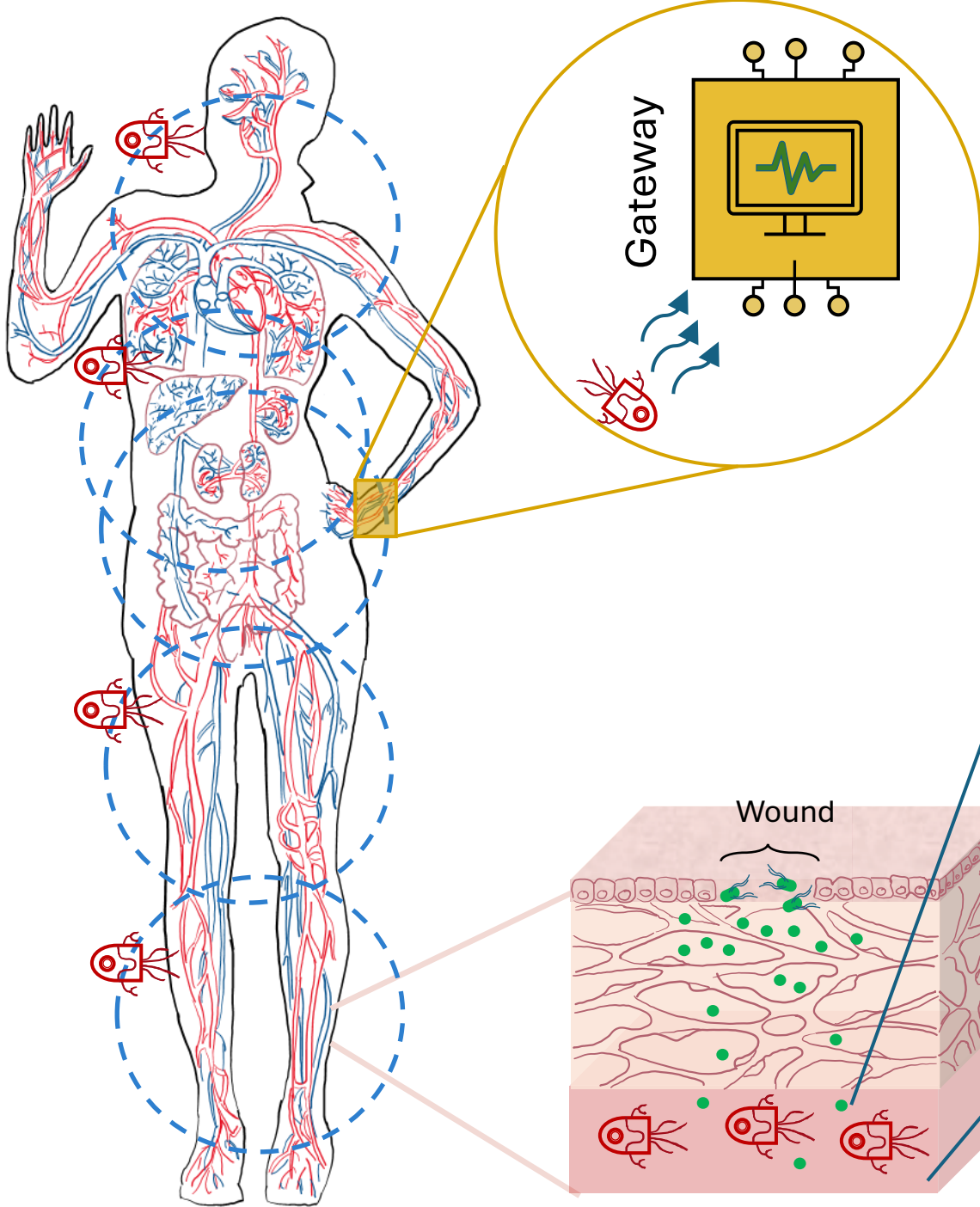
Associating Metabolic reactions with NN coefficients

Biological Networking: Physical Components

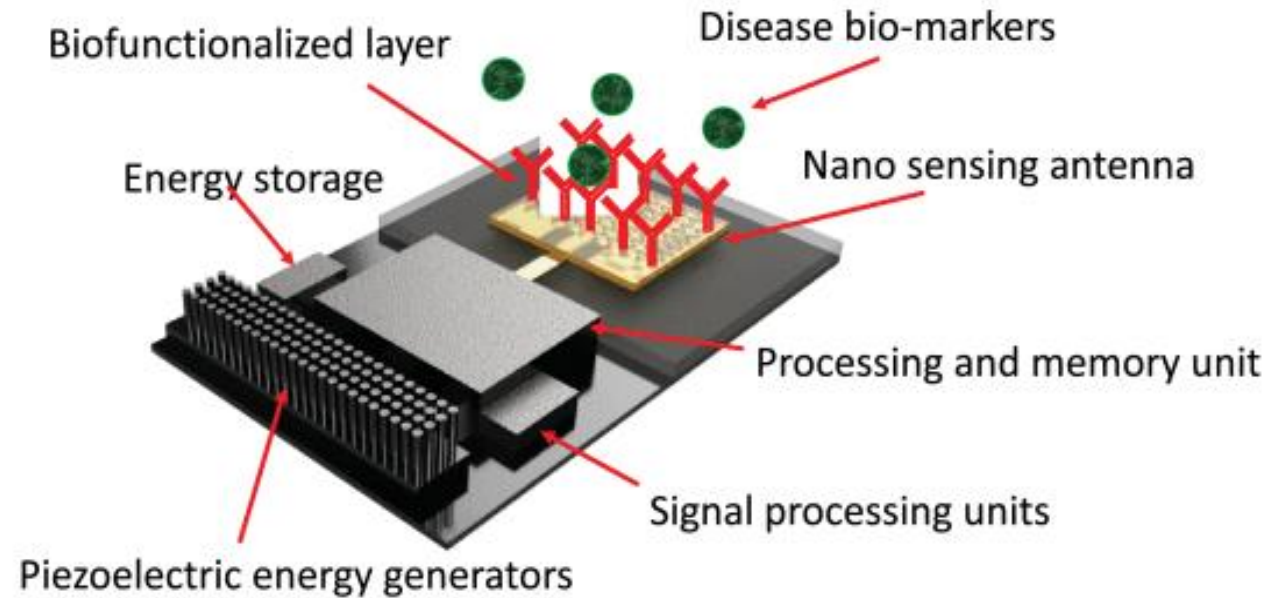




Internet of Bio-Nano-Things



Internet of Bio-Nano-Things



[1] A. Sangwan and J. M. Jornet, "Joint Communication and Bio-Sensing With Plasmonic Nano-Systems to Prevent the Spread of Infectious Diseases in the Internet of Nano-Bio Things," in *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 11, pp. 3271-3284, Nov. 2022, doi: 10.1109/JSAC.2022.3214288.

Bio-Building Carriers

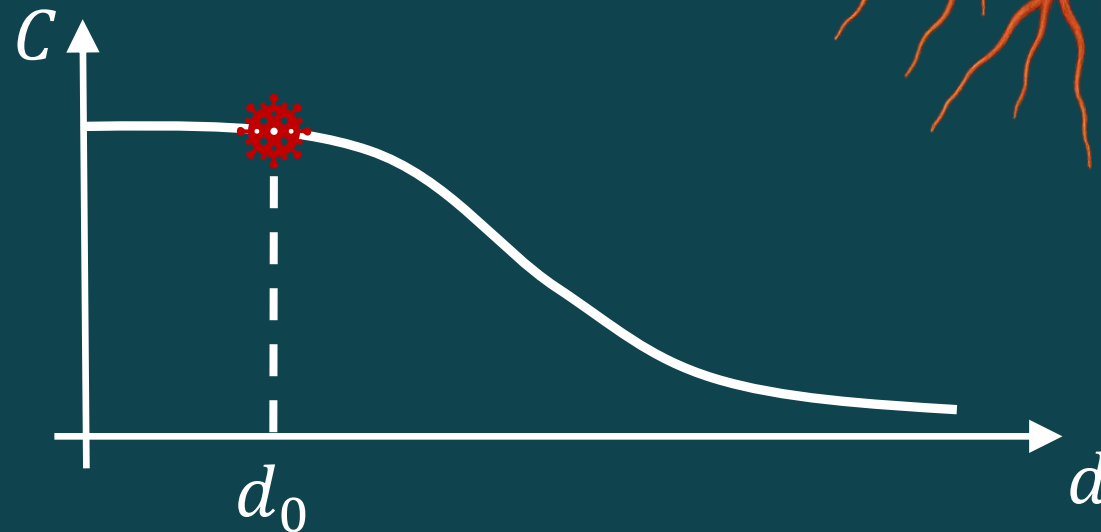
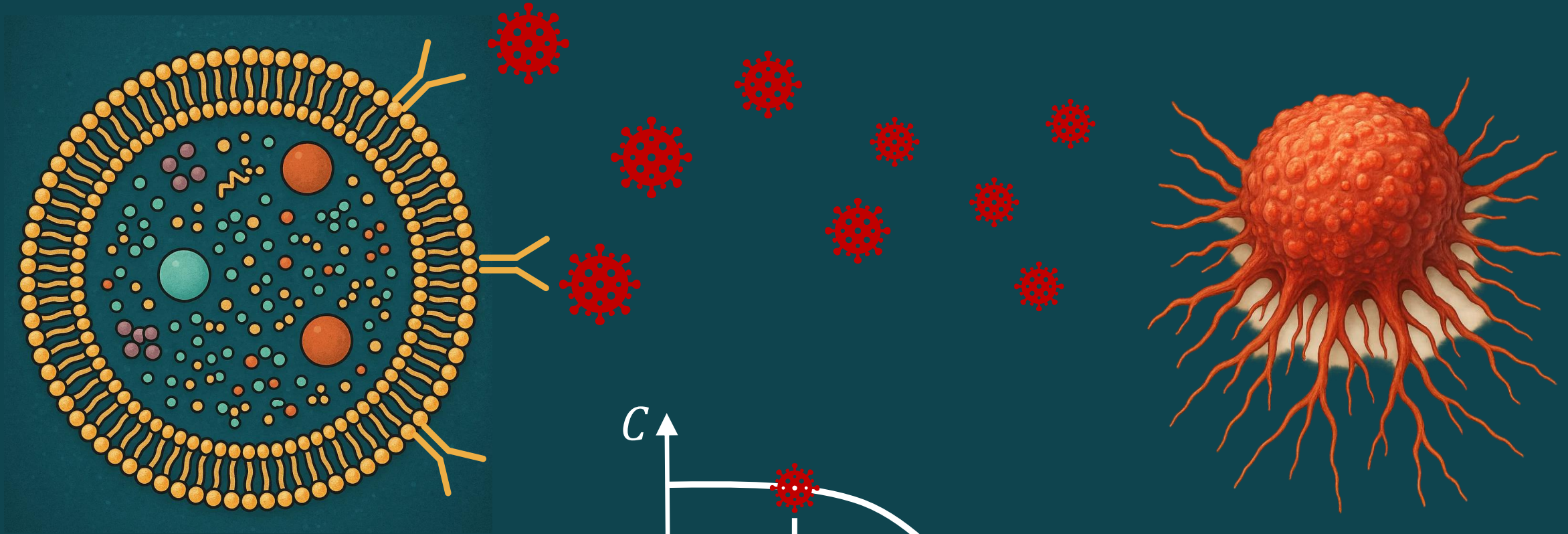


Cancer Cell



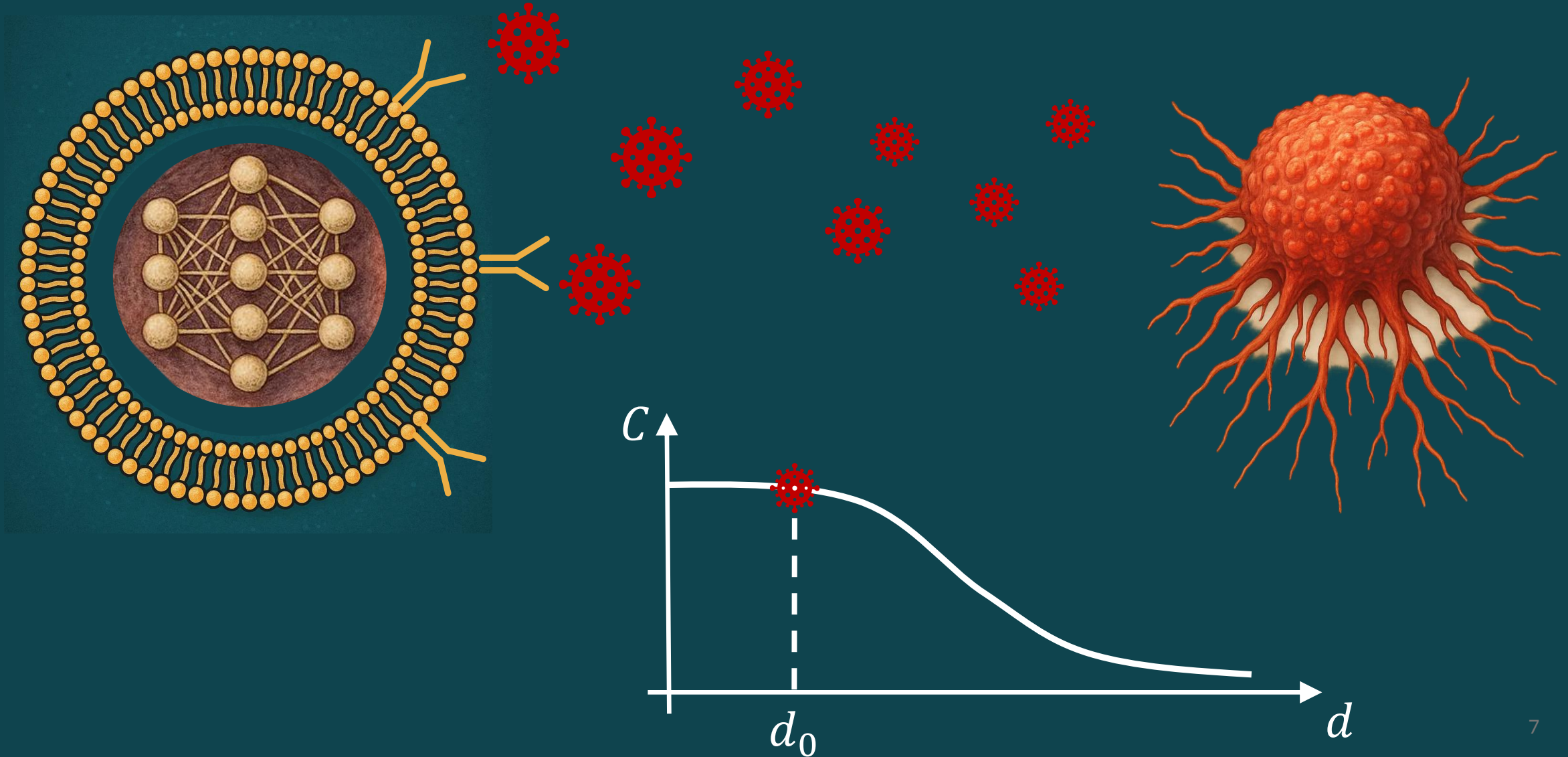
Bio-Building Carriers

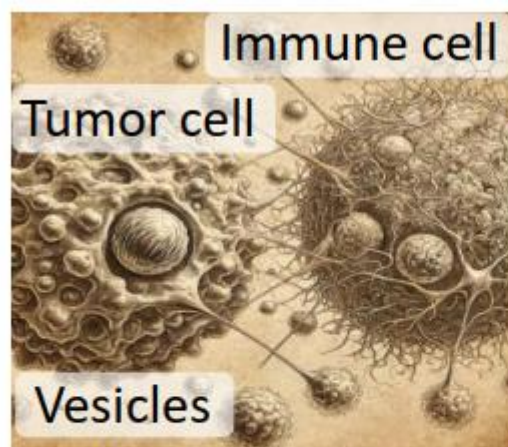
Cancer Cell



Bio-Building Carriers

Cancer Cell





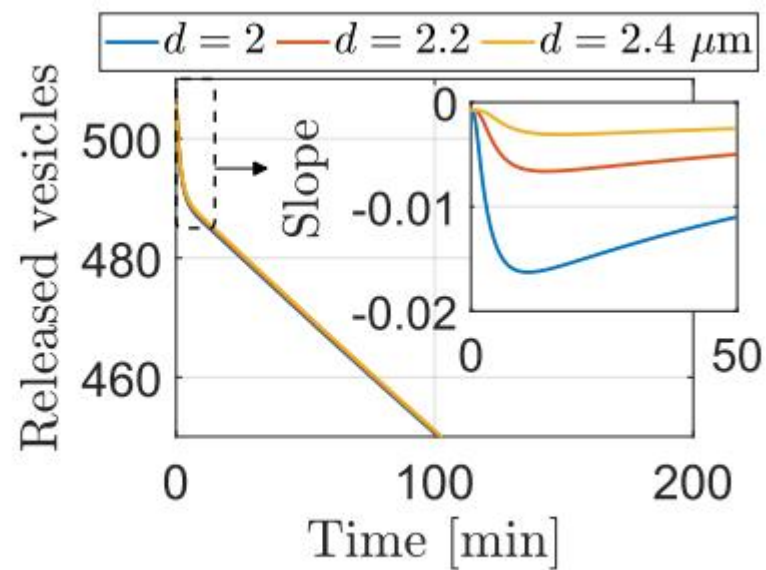
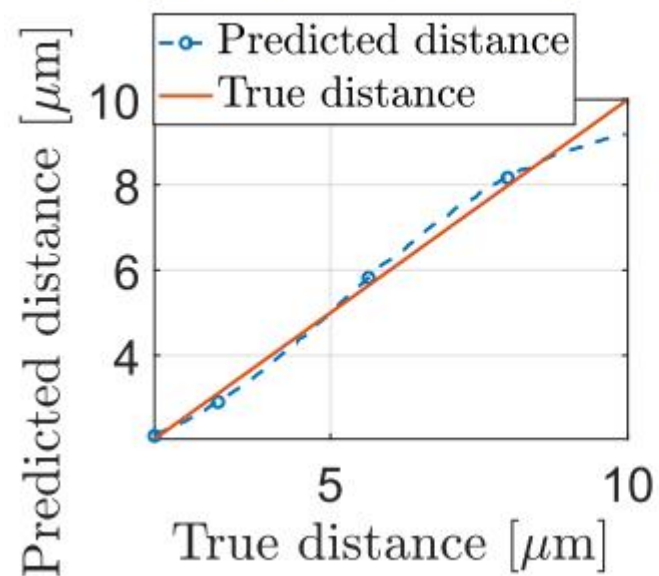
(a) Dataset

Feature
Extraction

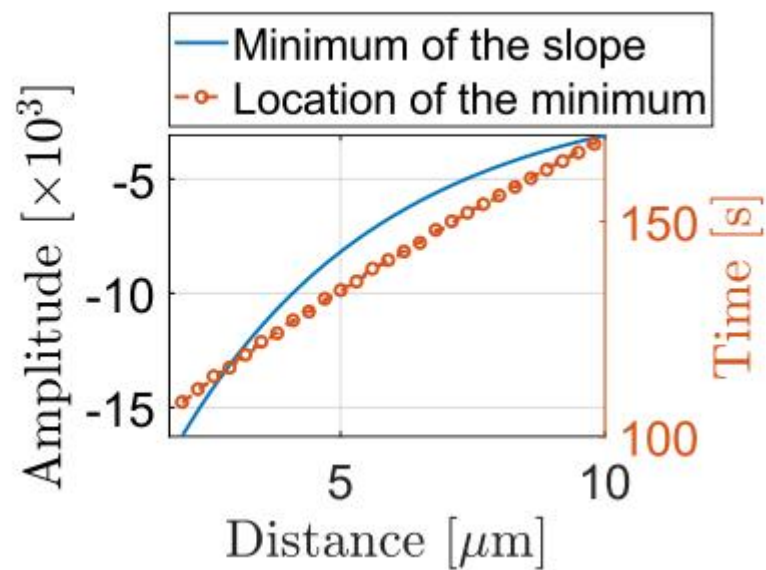
Neural
Network



(c) Estimated distance

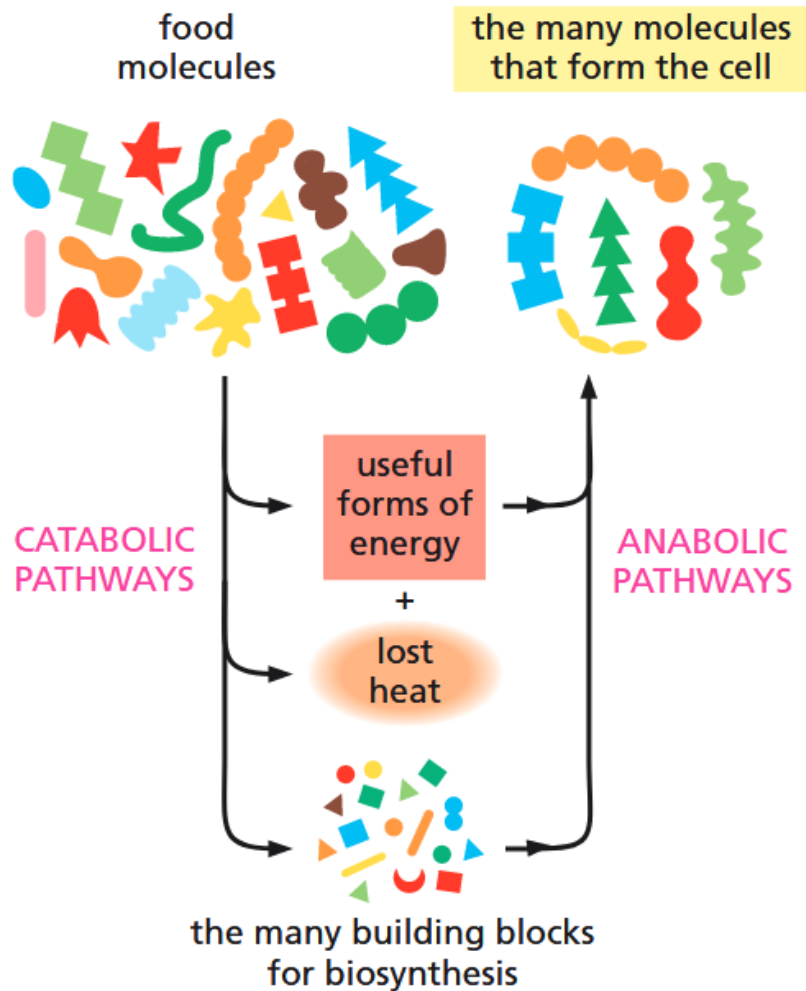


(b) Features



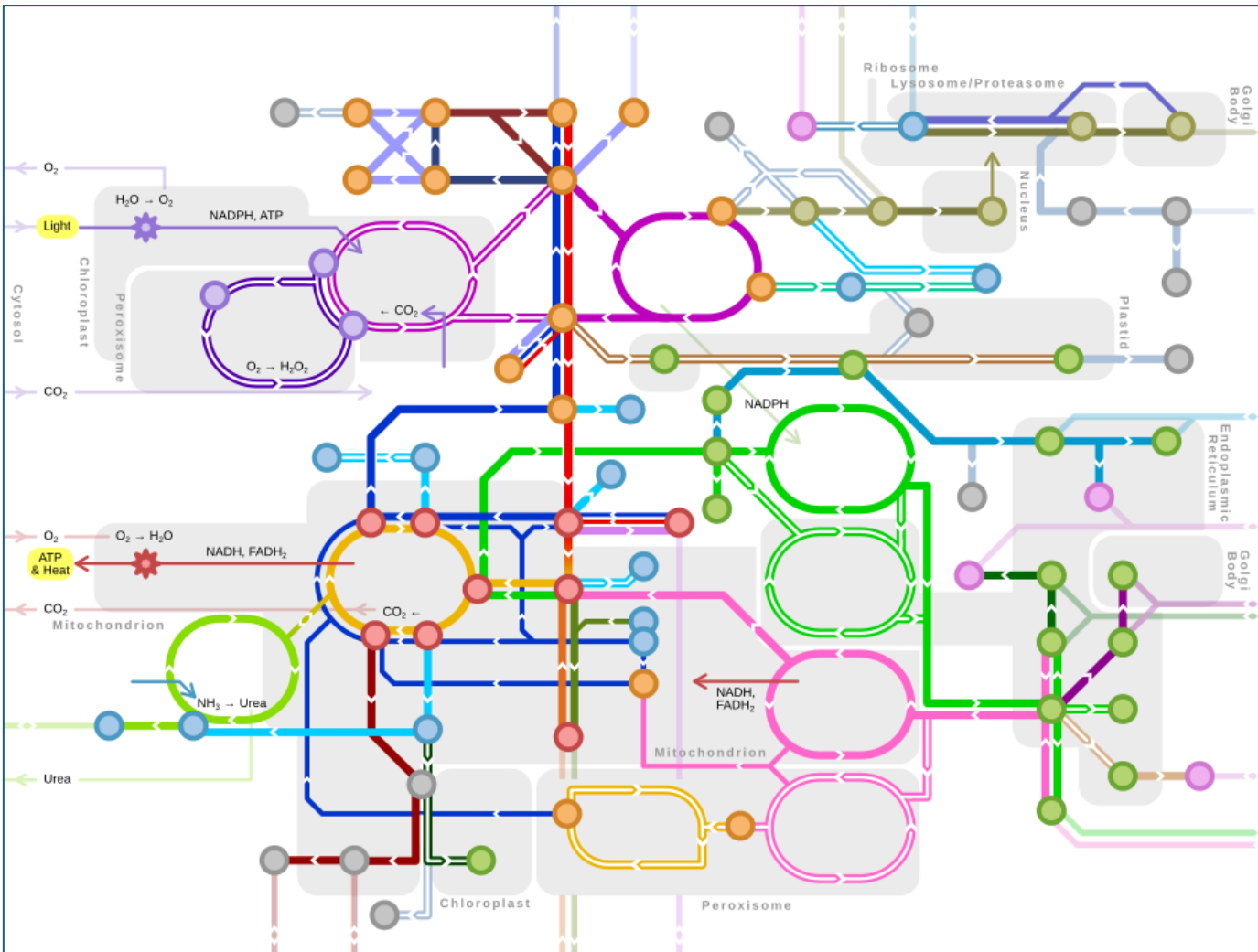
Concept

Metabolic Network

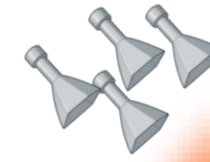


Concept

Metabolic Network



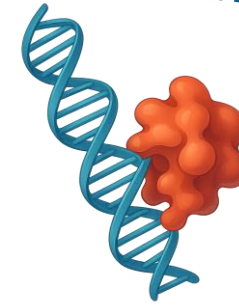
Antenna Array



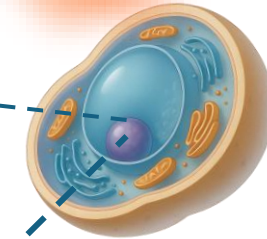
Terahertz Waveforms



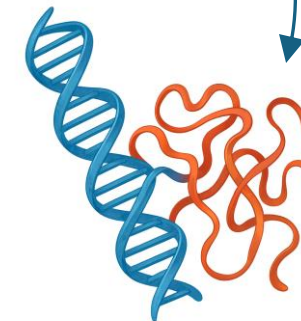
Folded Enzyme



Yeast Cell



Unfolded Enzyme



- Enzyme unfolding probability

$$p_U = 1 - \frac{1}{1 + e^{\left(\frac{\Delta E - \langle E_{tot} \rangle_{ss}}{k_b T}\right)}}$$

$\langle E_{tot} \rangle_{ss}$ Total energy of the enzyme

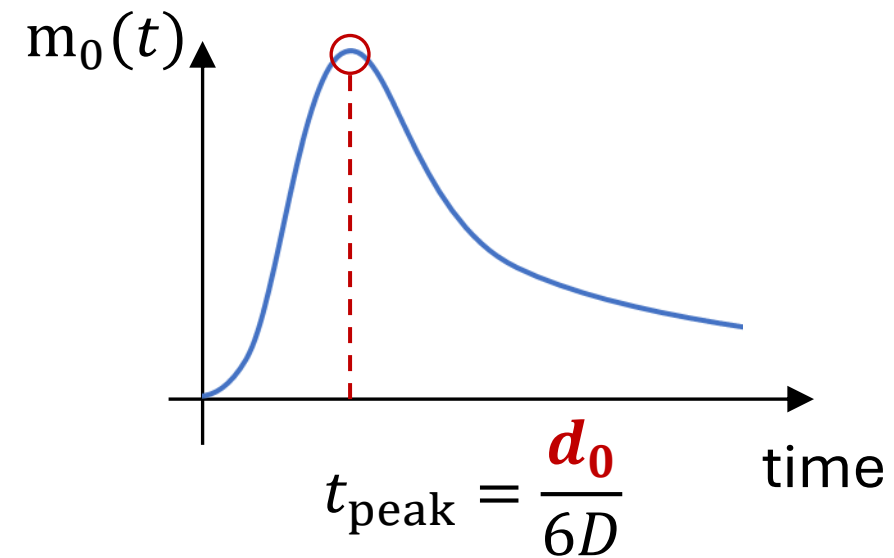
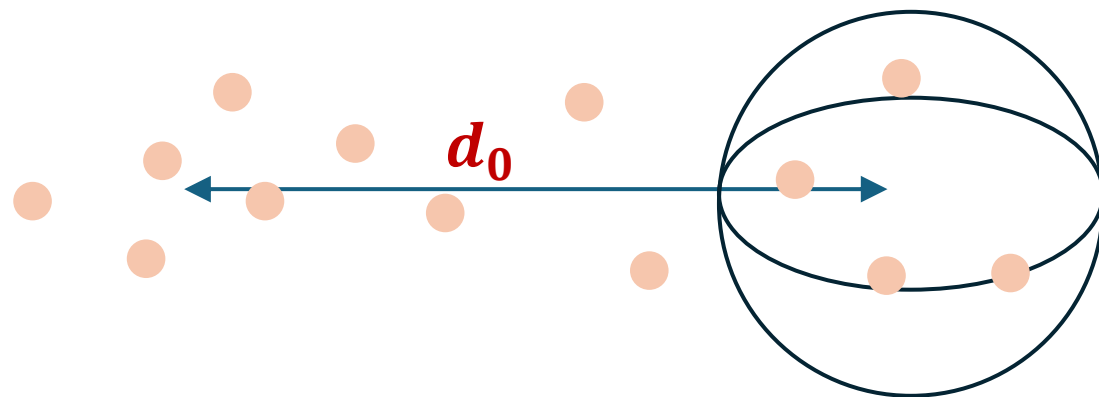
ΔE Enzyme-free energy

k_b Boltzmann constant

T Temperature of the Cell

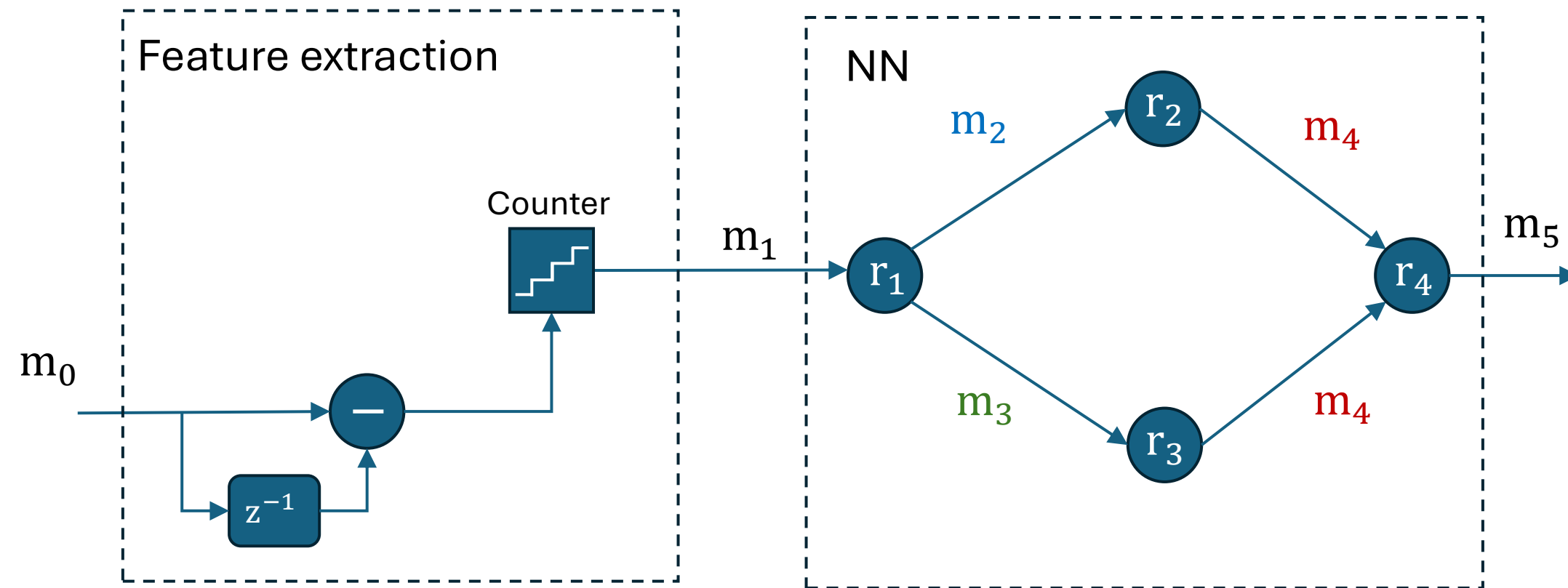
Point Transmitter

Receiver



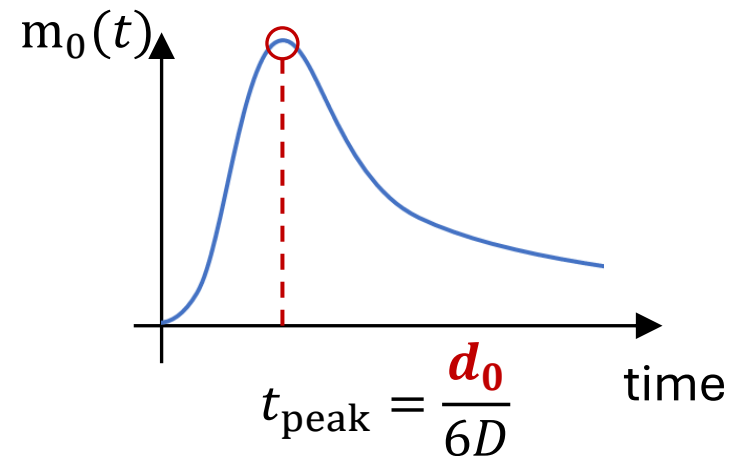
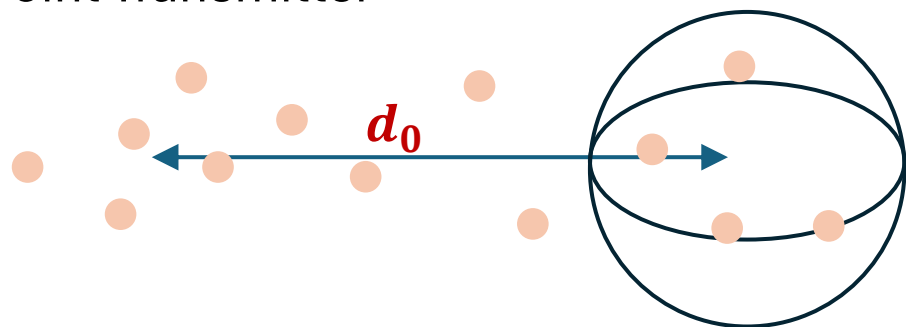
Feature extraction

NN



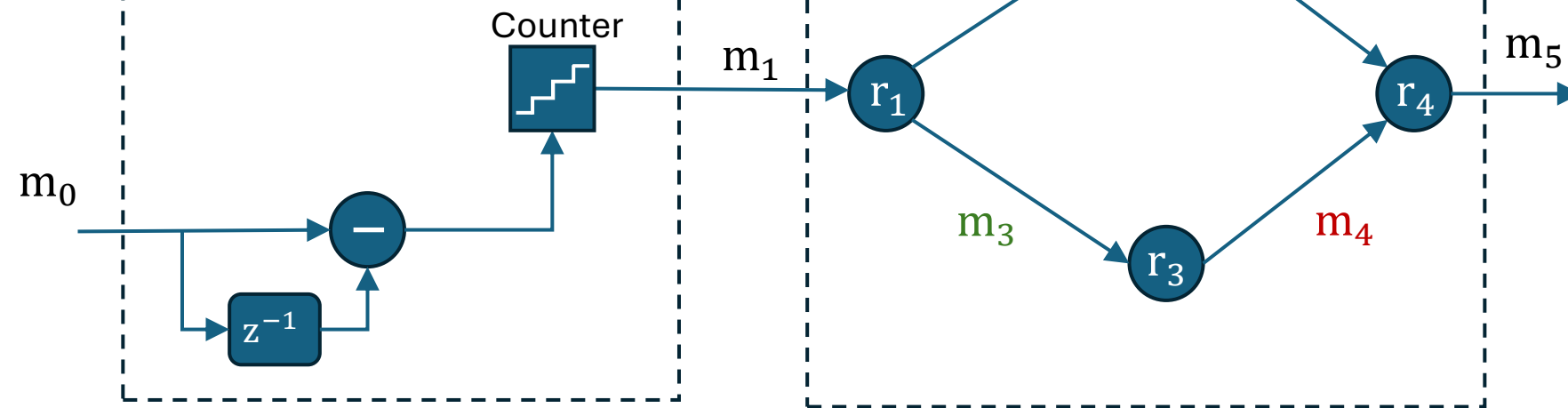
Point Transmitter

Receiver



Feature extraction

NN

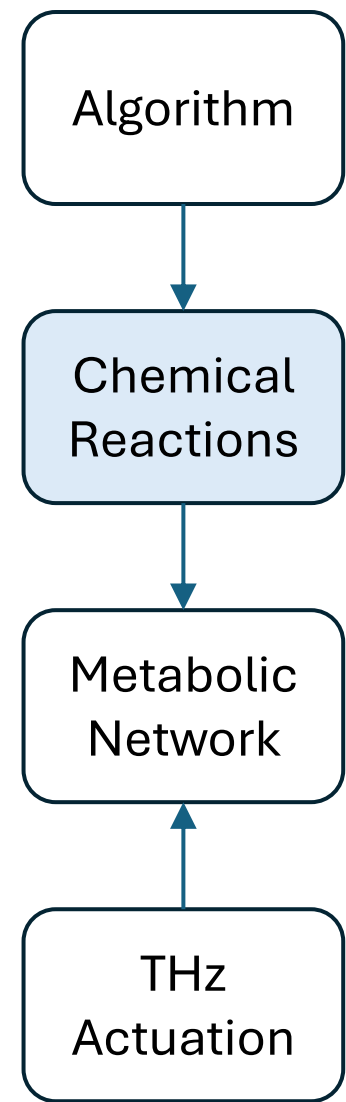
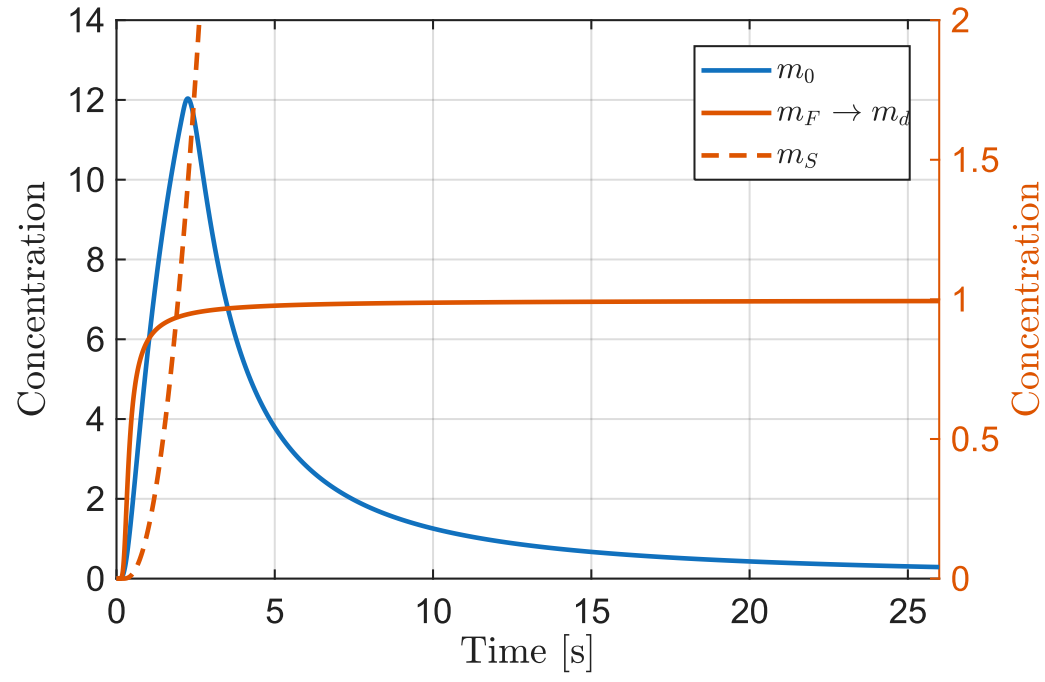
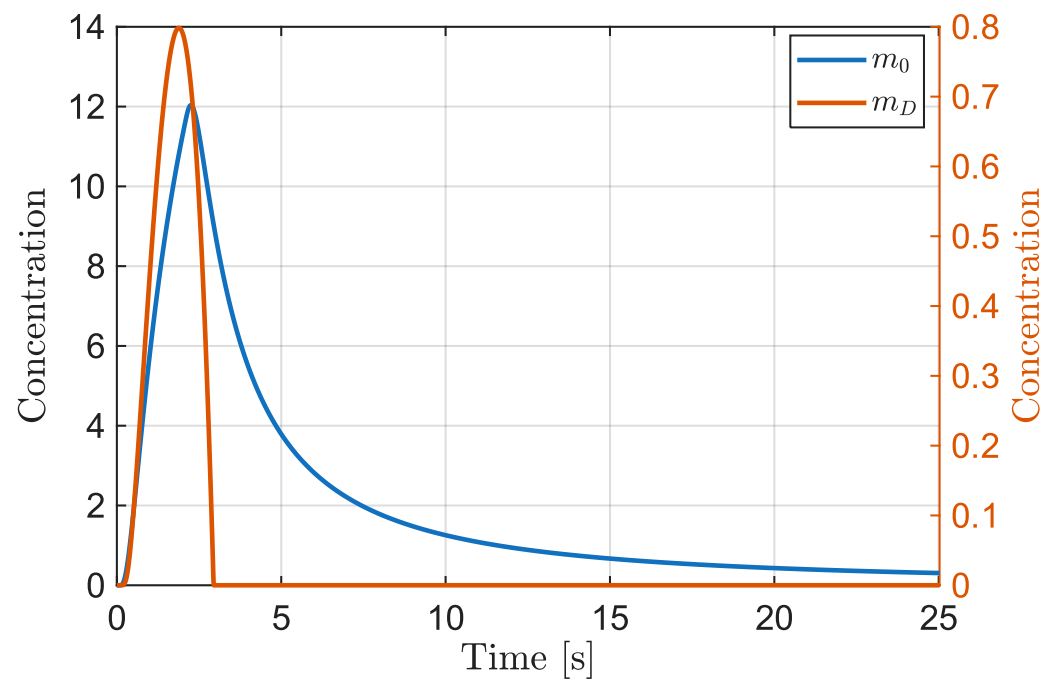
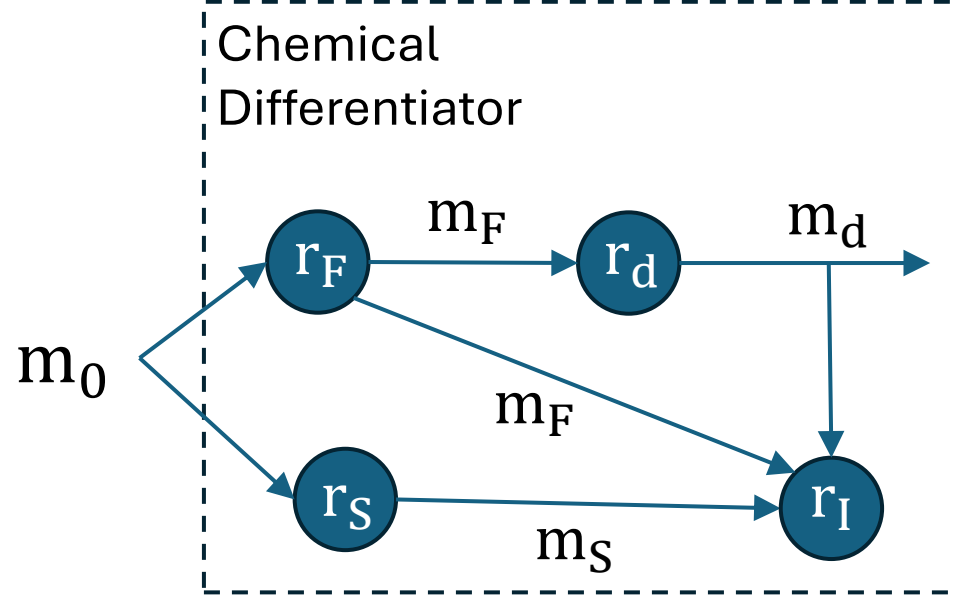
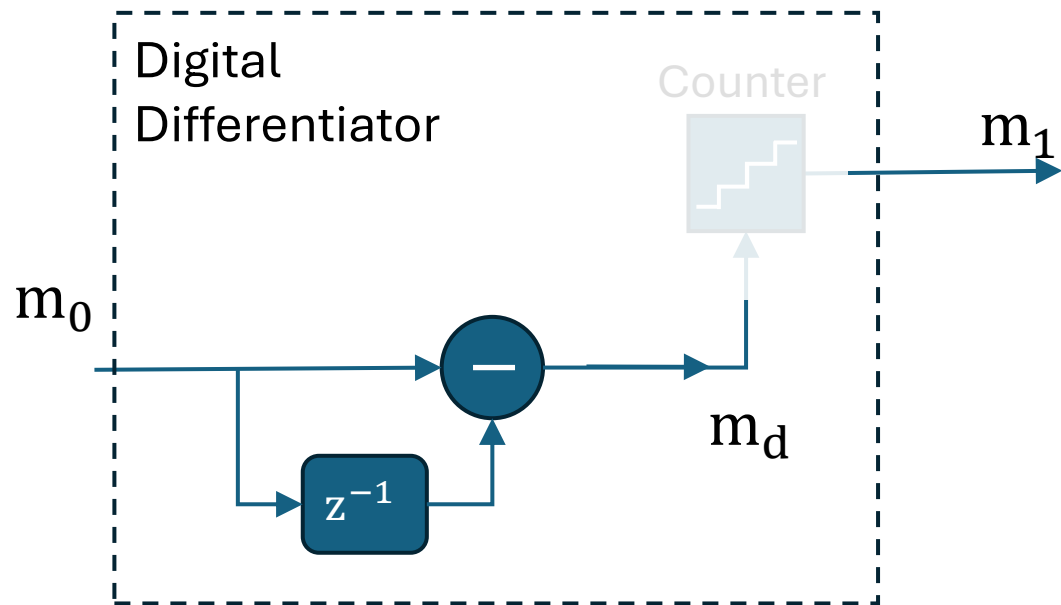


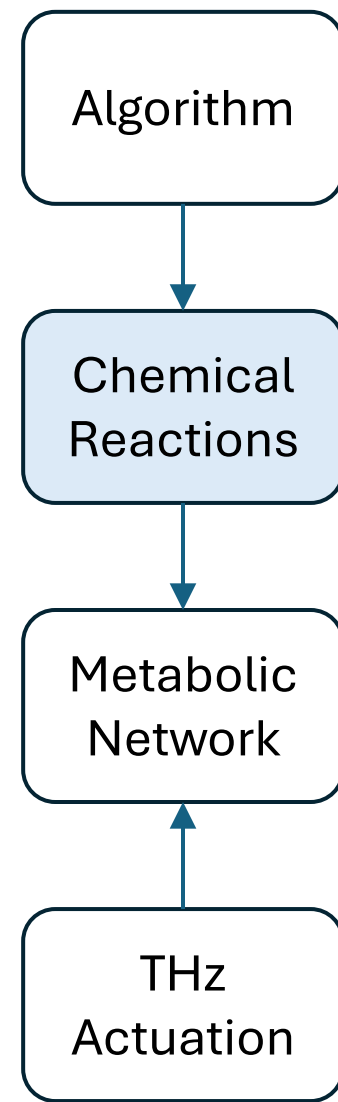
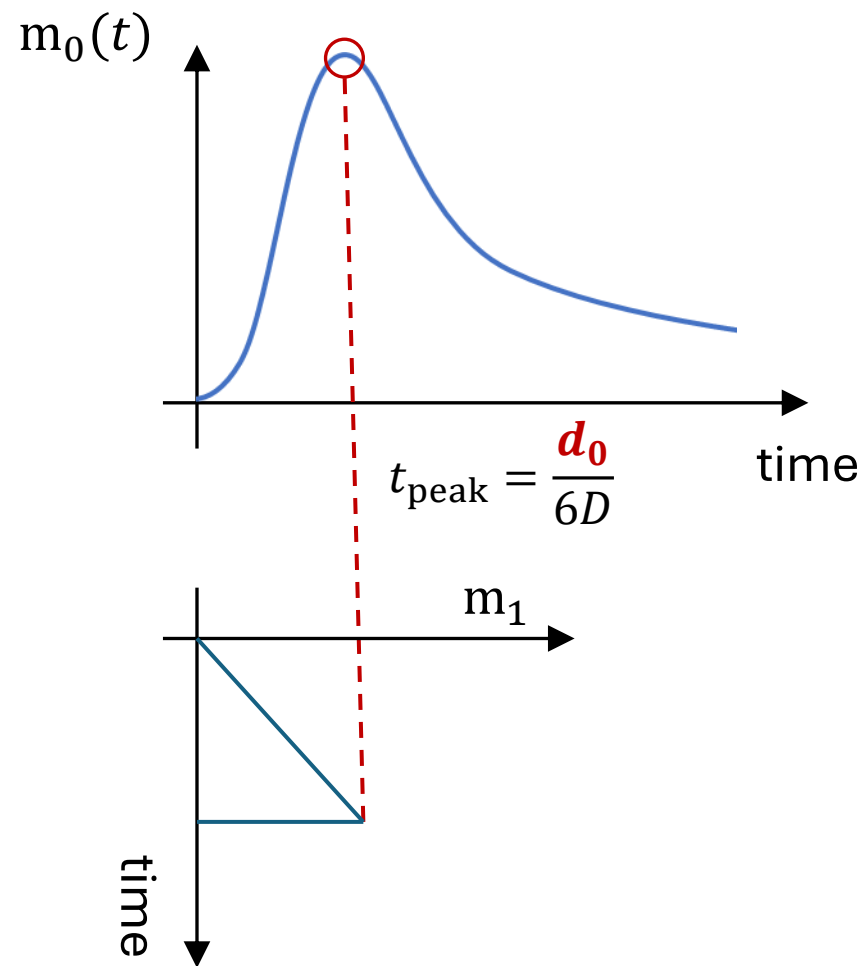
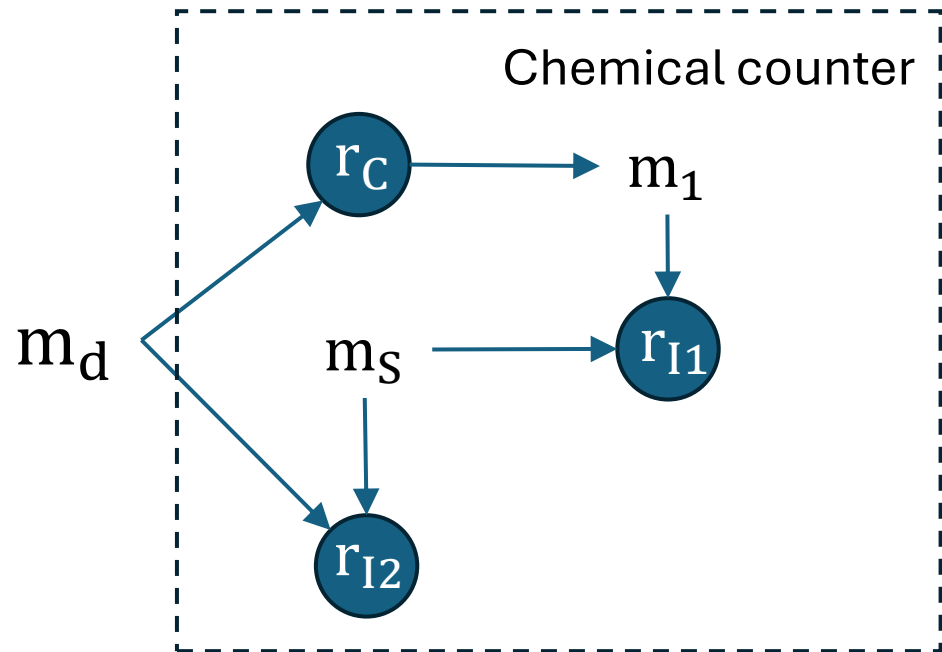
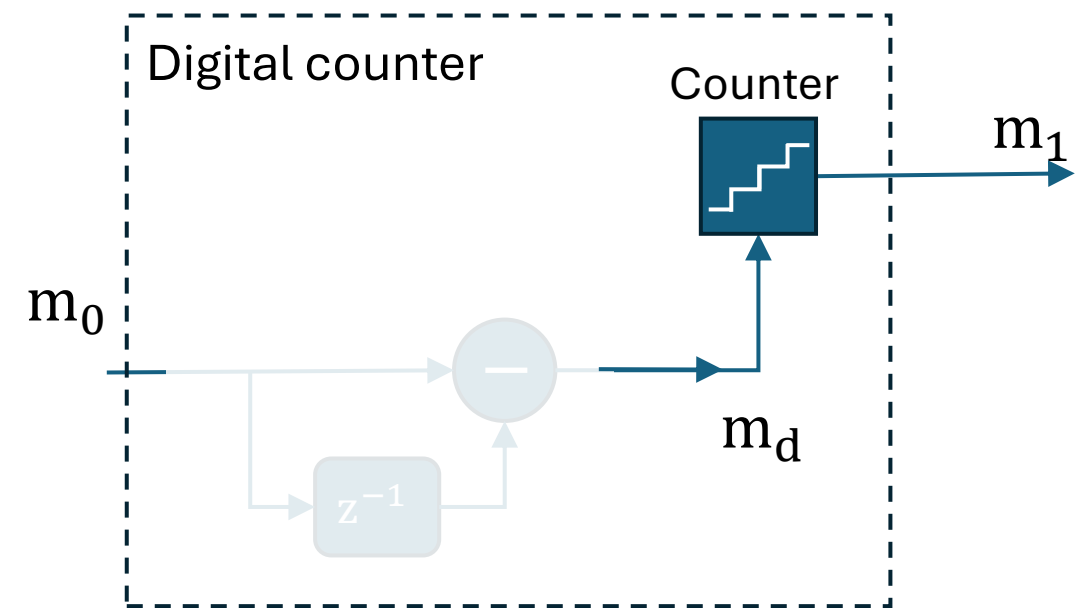
Algorithm

Chemical
Reactions

Metabolic
Network

THz
Actuation





Stoichiometric matrix

$$S_{\text{mets} \times \text{rxns}} = \begin{bmatrix} b_{11} & \cdots & b_{1,\text{rxn}} \\ \vdots & \ddots & \vdots \\ b_{\text{mets},1} & \cdots & b_{\text{mets},\text{rxns}} \end{bmatrix}, b_{ij} \in \mathbb{Z} \quad (1)$$

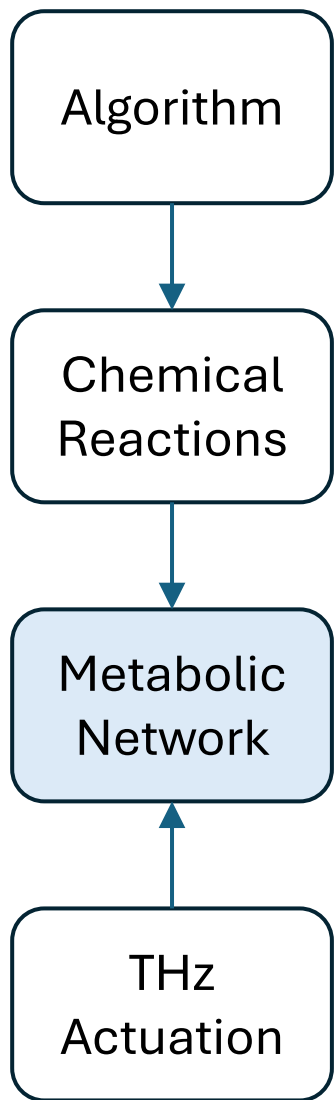
Example:

Reaction r_1 : $m_1 + 2m_2 \rightarrow m_3$ (2)

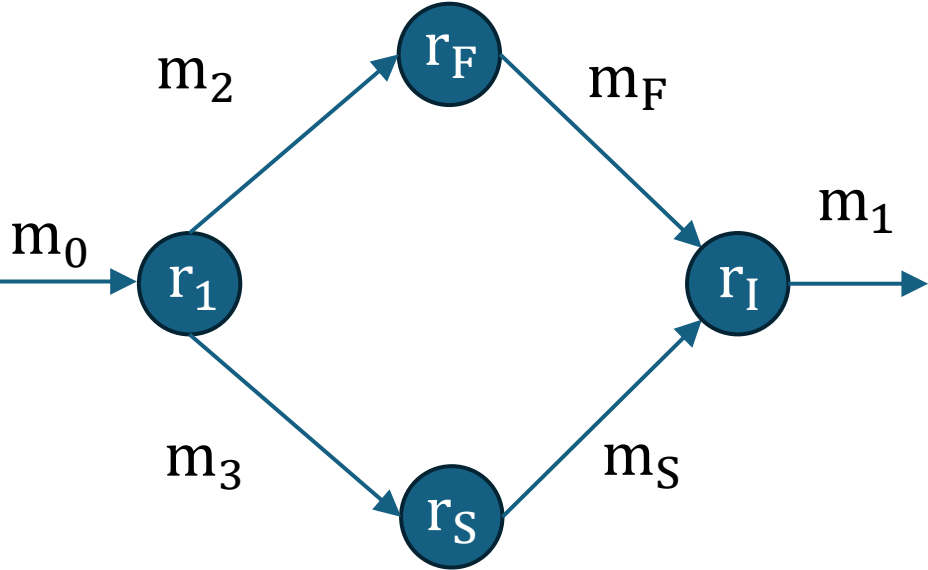
$$b_{11} = -1,$$

$$b_{21} = -2,$$

$$b_{31} = 1$$

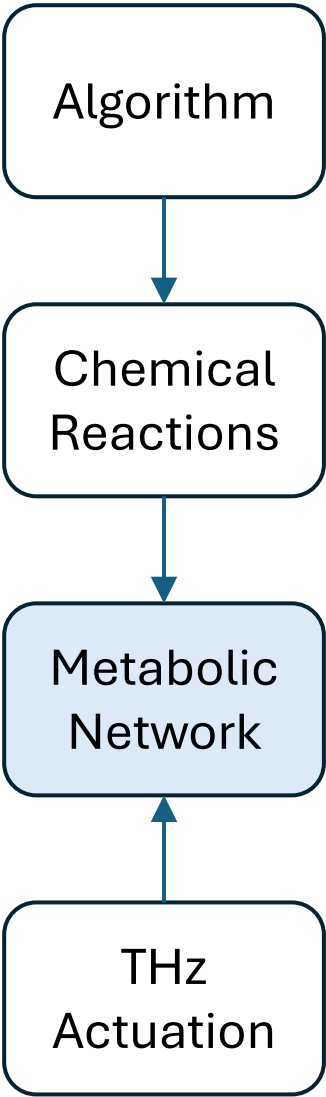


Mapping a 2-neuron NN

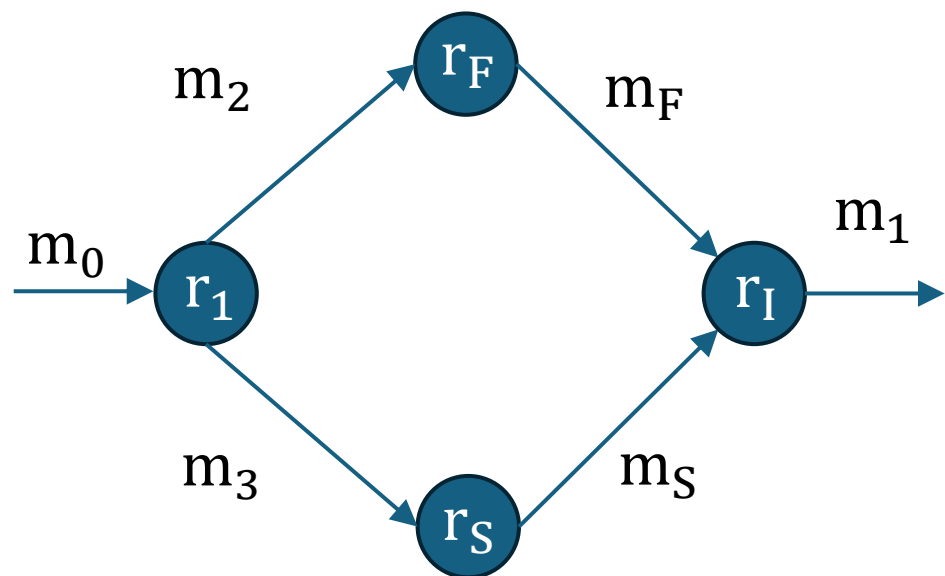


Pattern

	r_1	r_I	r_F	r_S	r_5
m_0	-				
m_1		+			
m_2	+		-		
m_3	+			-	
m_F		-	+		
m_S		-		+	



Solutions



Reaction 1: peroxisomal acyl-CoA thioesterase

EC code: 3.1.2.2

PBD ID: 1TUB

Gene: YJR019C

Formula:

$C_{31}H_{50}N_7O_{17}P_3S + H_2O \rightarrow C_{21}H_{32}N_7O_{16}P_3S + C_{10}H_{19}O_2 + H$

Location: Peroxisome

Reaction F: acetyl-CoA C-acyltransferase (decanoyl-CoA)

EC code: 2.3.1.16

Gene: YIL160C

$C_{33}H_{52}N_7O_{18}P_3S + C_{21}H_{32}N_7O_{16}P_3S \rightarrow C_{23}H_{34}N_7O_{17}P_3S + C_{31}H_{50}N_7O_{17}P_3S$

Location: Peroxisome

Reaction S: fatty-acid--CoA ligase (decanoate)

EC code: 6.2.1.3

Gene: YER015W

$C_{10}H_{12}N_5O_{13}P_3 + C_{21}H_{32}N_7O_{16}P_3S + C_{10}H_{19}O_2 \rightarrow C_{10}H_{12}N_5O_7P + C_{31}H_{50}N_7O_{17}P_3S + HO_7P_2$

Location: Peroxisome

Algorithm

Chemical
Reactions

Metabolic
Network

THz
Actuation

Enzyme unfolding probability

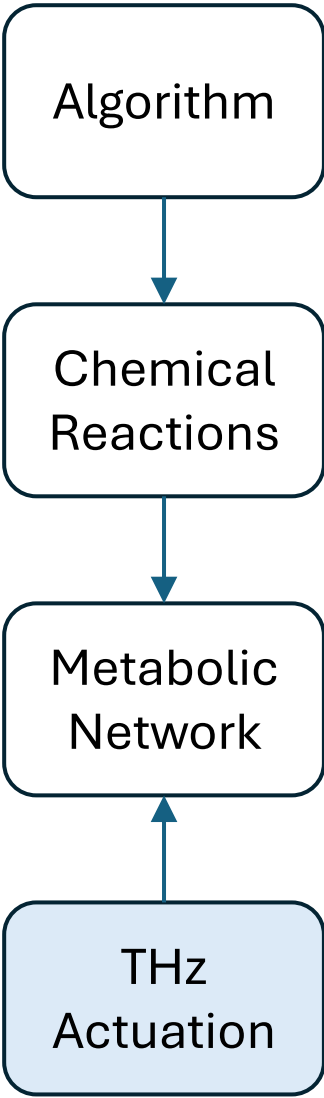
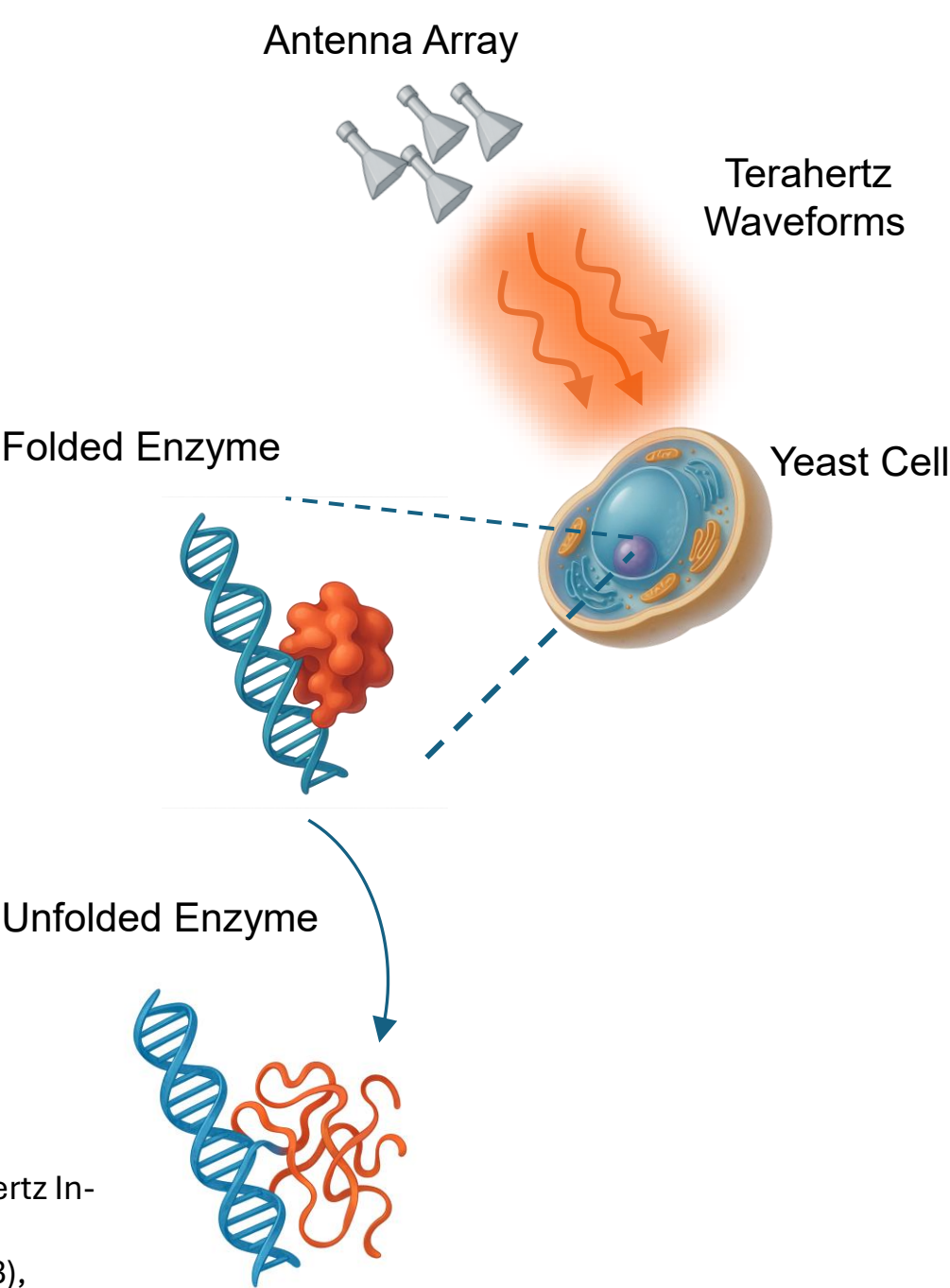
$$p_U = 1 - \frac{1}{1 + e^{\left(\frac{\Delta E - \langle E_{tot} \rangle_{ss}}{k_b T}\right)}}$$

$\langle E_{tot} \rangle_{ss}$ Total energy of the enzyme

ΔE Enzyme-free energy

k_b Boltzmann constant

T Temperature of the Cell

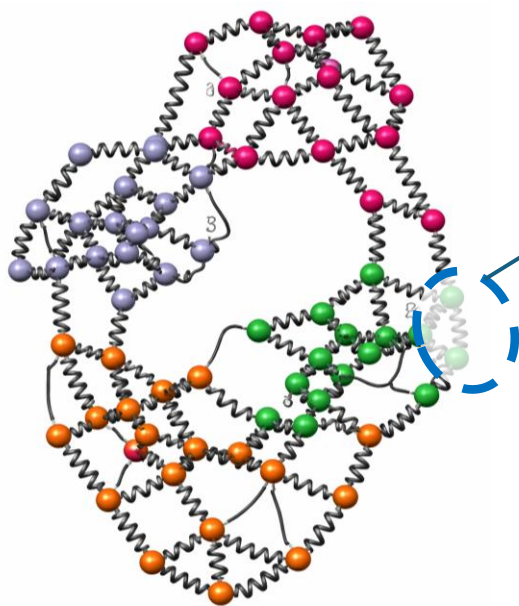


[1] Hadeel Elayan, Andrew W. Eckford, and Raviraj S. Adve. 2023. Terahertz Induced Protein Interactions in a Random Medium. IEEE Transactions on Molecular, Biological and Multi-Scale Communications 9, 4 (Dec. 2023), 435–446. <https://doi.org/10.1109/tmbmc.2023.3327302>

Enzyme
PBD ID: 1TUB



Anisotropic
Network
Model



Langevin Equation

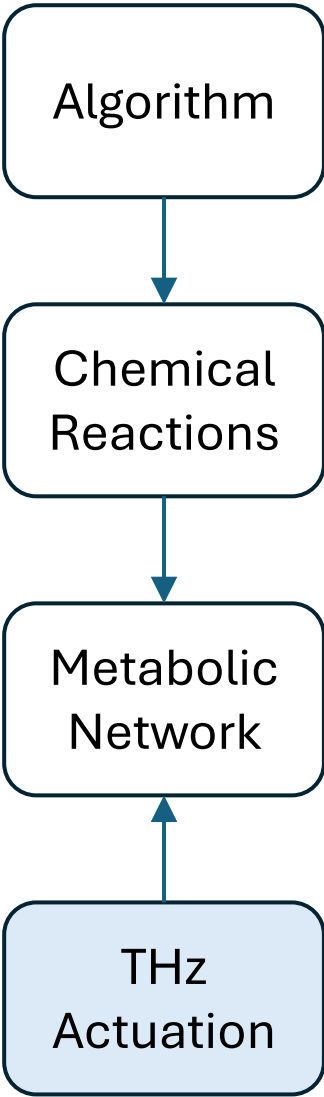
$$\mathbf{M}\ddot{\mathbf{x}} = -\mathbf{H}\mathbf{x} - \mathbf{B}\dot{\mathbf{x}} + \mathbf{f}_{\text{THz}}(t) + \boldsymbol{\xi}(t) \quad (1)$$

$$f_{\text{THz}}(t) = f_o \cos(\omega_a t + \Theta) \quad (2)$$

$$f_o = 2\pi r^3 \epsilon_m \text{Re}\{\mathcal{K}\} \nabla |\mathbf{E}_{\text{rms}}|^2 \quad (3)$$

$$p_U = 1 - \frac{1}{1 + e^{\left(\frac{\Delta E - \langle E_{\text{tot}} \rangle_{ss}}{k_b T}\right)}} \quad (4)$$

$$\langle E_{\text{tot}} \rangle = \frac{\alpha}{4\beta m [(\omega_o^2 - \omega_a^2)^2 + (\omega_a \gamma)^2]} (\omega_o^2 + \omega_a^2)^2 \quad (5)$$

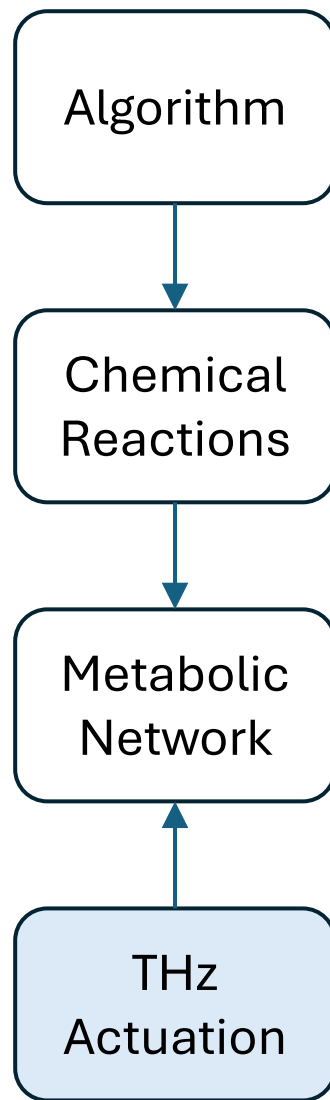
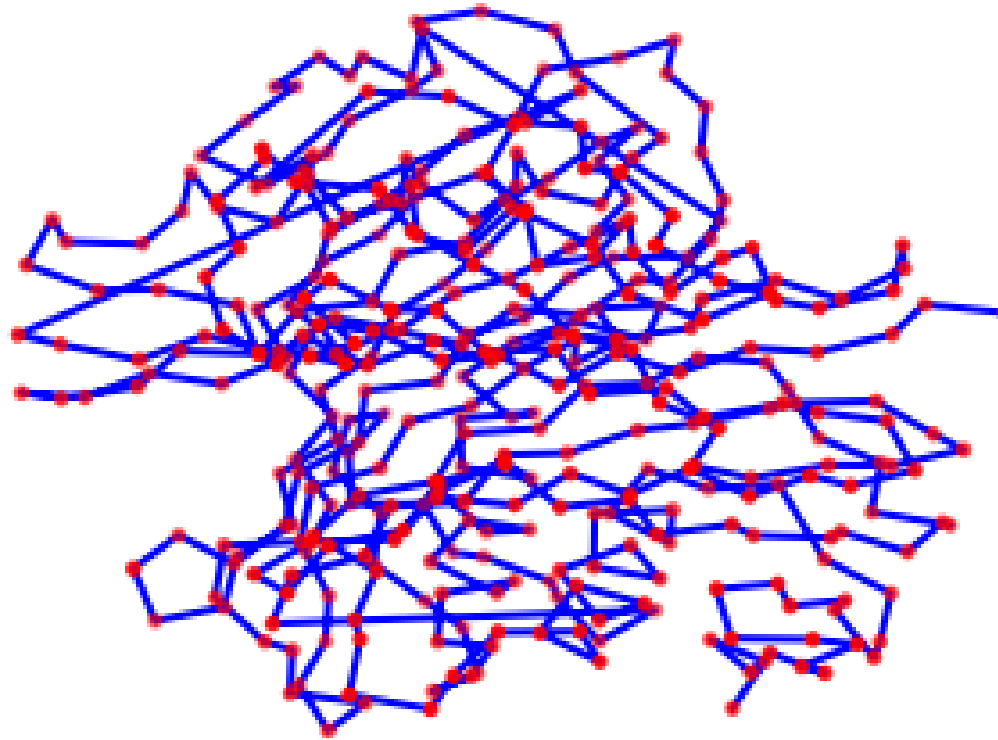


[1] Hadeel Elayan, Andrew W. Eckford, and Raviraj S. Adve. 2023. Terahertz Induced Protein Interactions in a Random Medium. IEEE Transactions on Molecular, Biological and Multi-Scale Communications 9, 4 (Dec. 2023), 435–446. <https://doi.org/10.1109/tmbmc.2023.3327302>

Enzyme PDB ID: 1TUB



Anisotropic Network Model (ANM)



Fundamentals of Flux Balance Analysis (FBA)

Stoichiometric matrix:

$$S_{\text{mets} \times \text{rxns}} = \begin{bmatrix} b_{11} & \cdots & b_{1,\text{rxn}} \\ \vdots & \ddots & \vdots \\ b_{\text{mets},1} & \cdots & b_{\text{mets},\text{rxns}} \end{bmatrix}, b_{ij} \in \mathbb{Z} \quad (3)$$

Reaction flux:

$$v \in \mathbb{R}^{|\text{rxns}|} \quad (4)$$

Example:

Reaction r_1 : $m_1 + 2m_2 \rightarrow m_3$

$$b_{11} = -1,$$

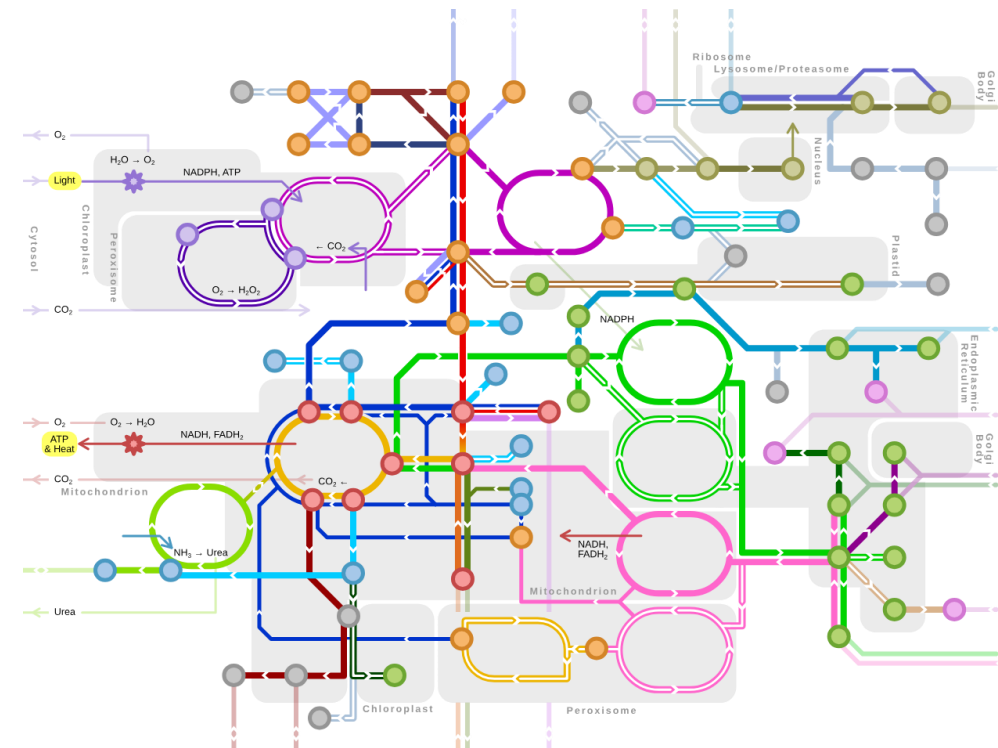
$$b_{21} = -2,$$

$$b_{31} = 1$$

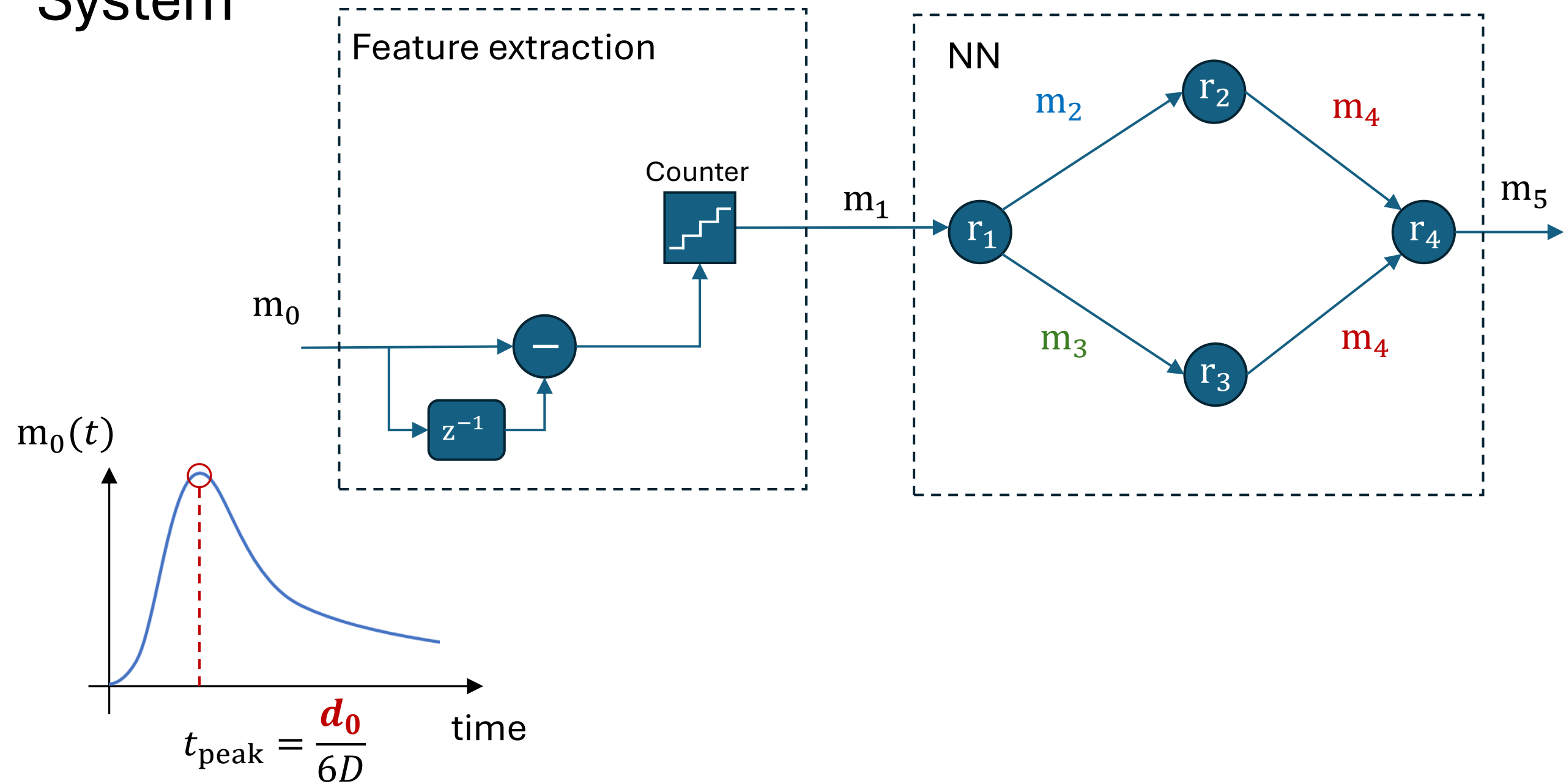
$$\frac{dx_m}{dt} = \sum_r S_{m,r} v_r = 0 \quad (1)$$

$\Downarrow$

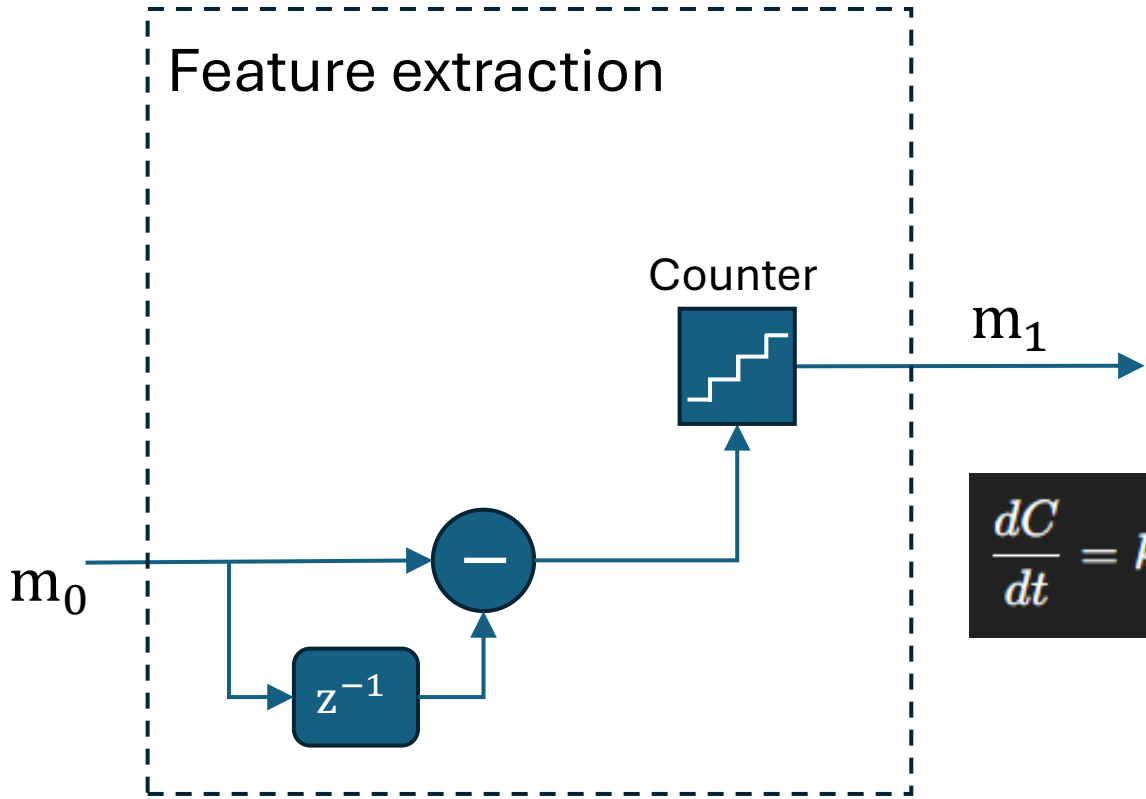
$$\begin{aligned} & \max_v c^T v \\ & \text{s.t. } Sv = 0 \\ & v \in [l \ u] \end{aligned} \quad (2)$$



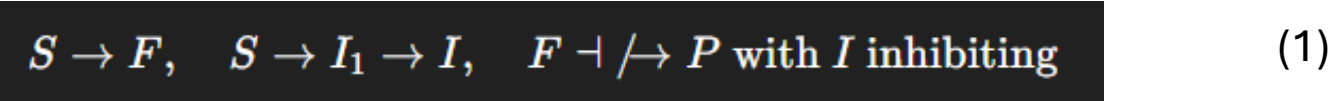
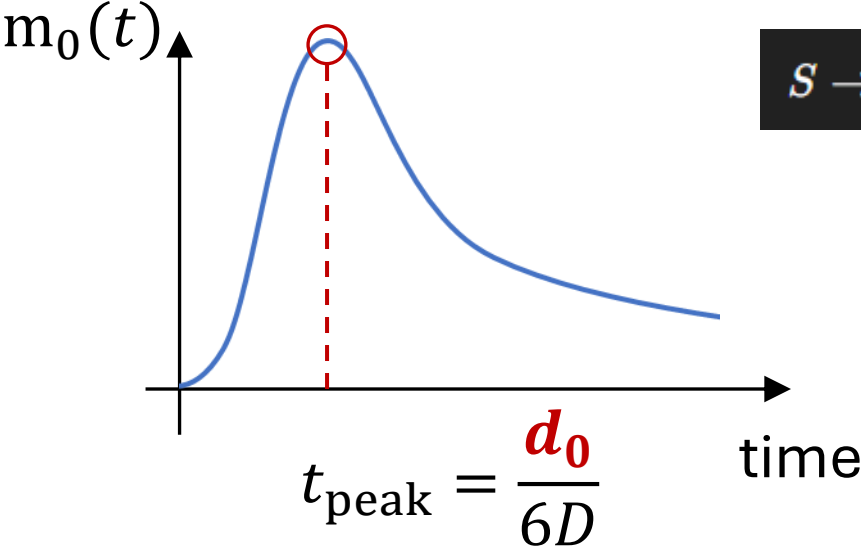
System

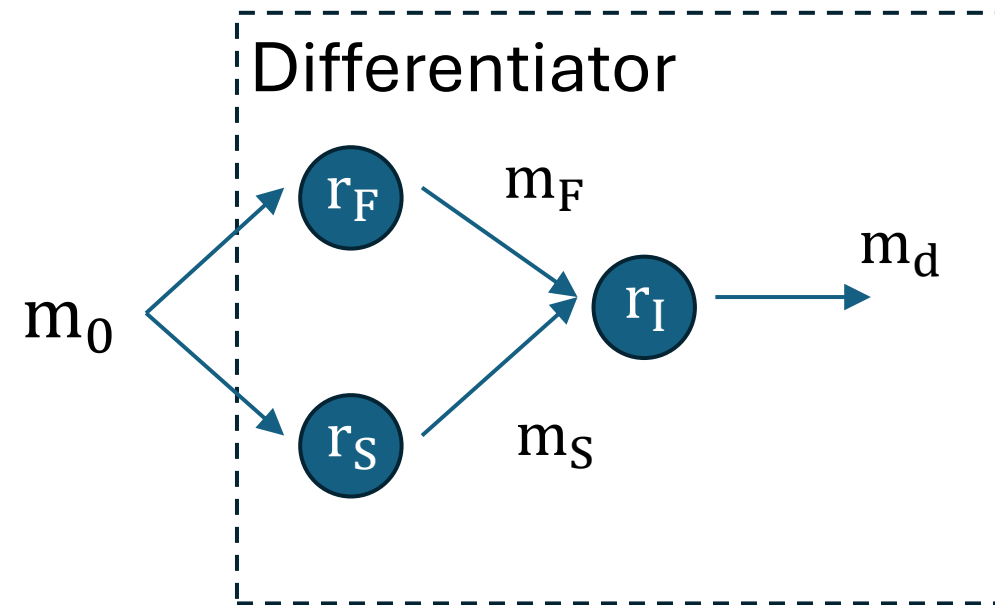
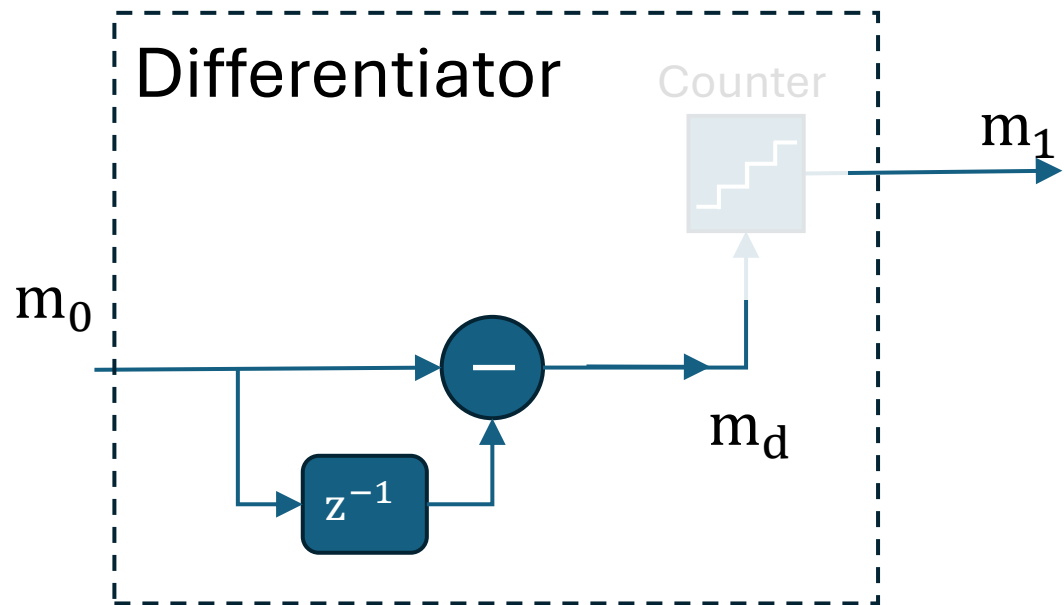


System



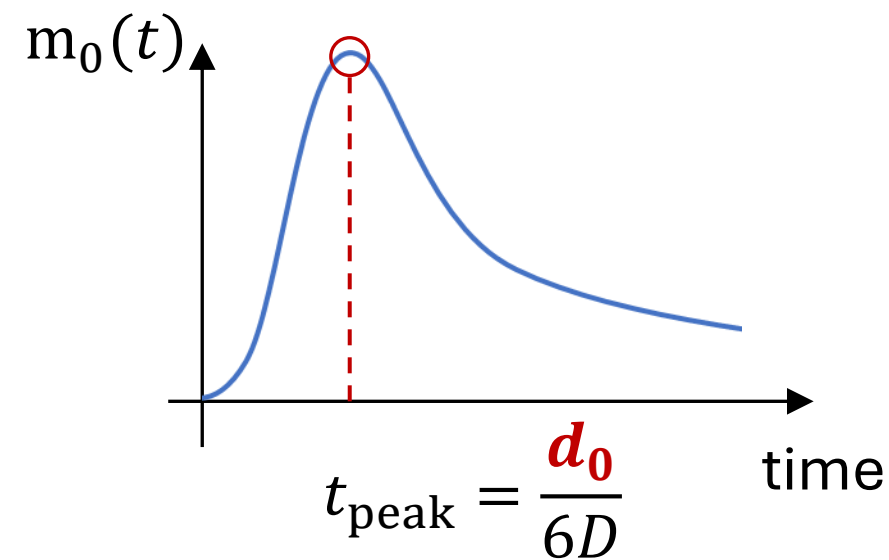
$$\frac{dC}{dt} = k_{\text{count}} \frac{P^n}{K_c^n + P^n} - k_{\text{reset}} \frac{K_r^m}{K_r^m + P^m} C \quad (2)$$

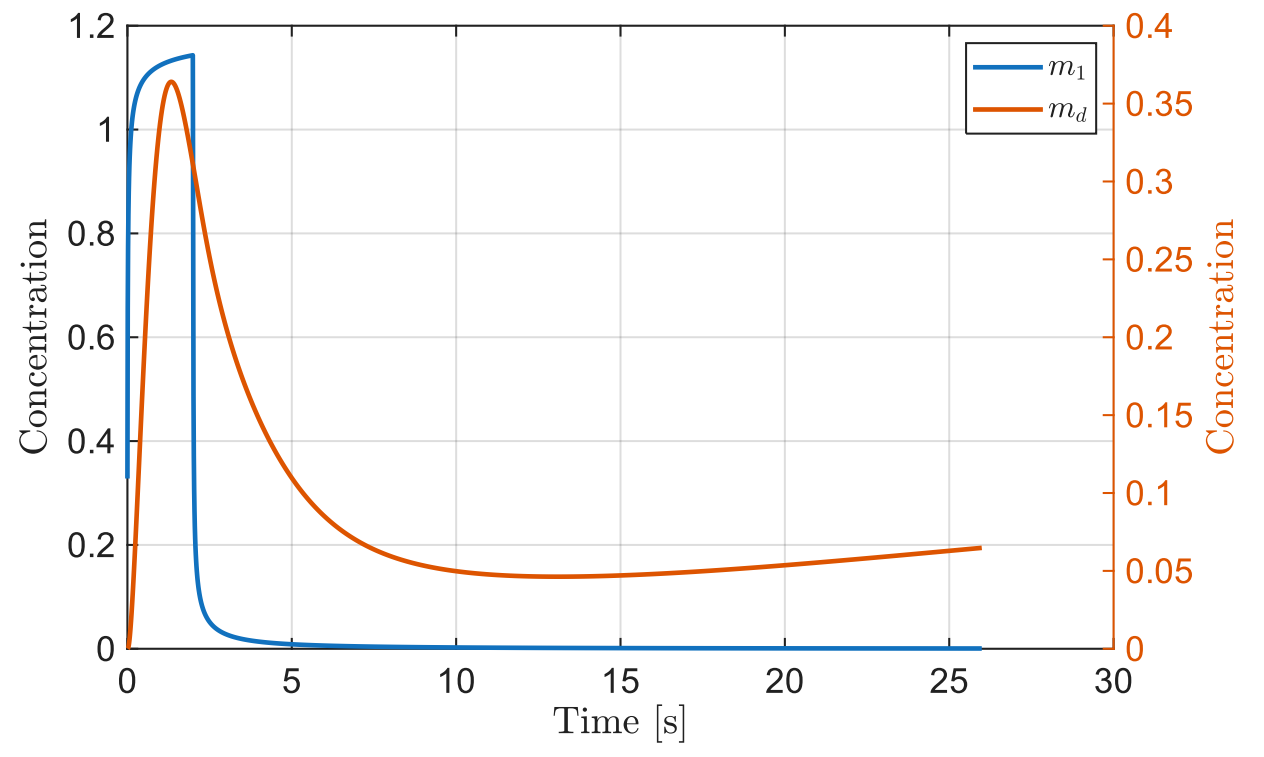
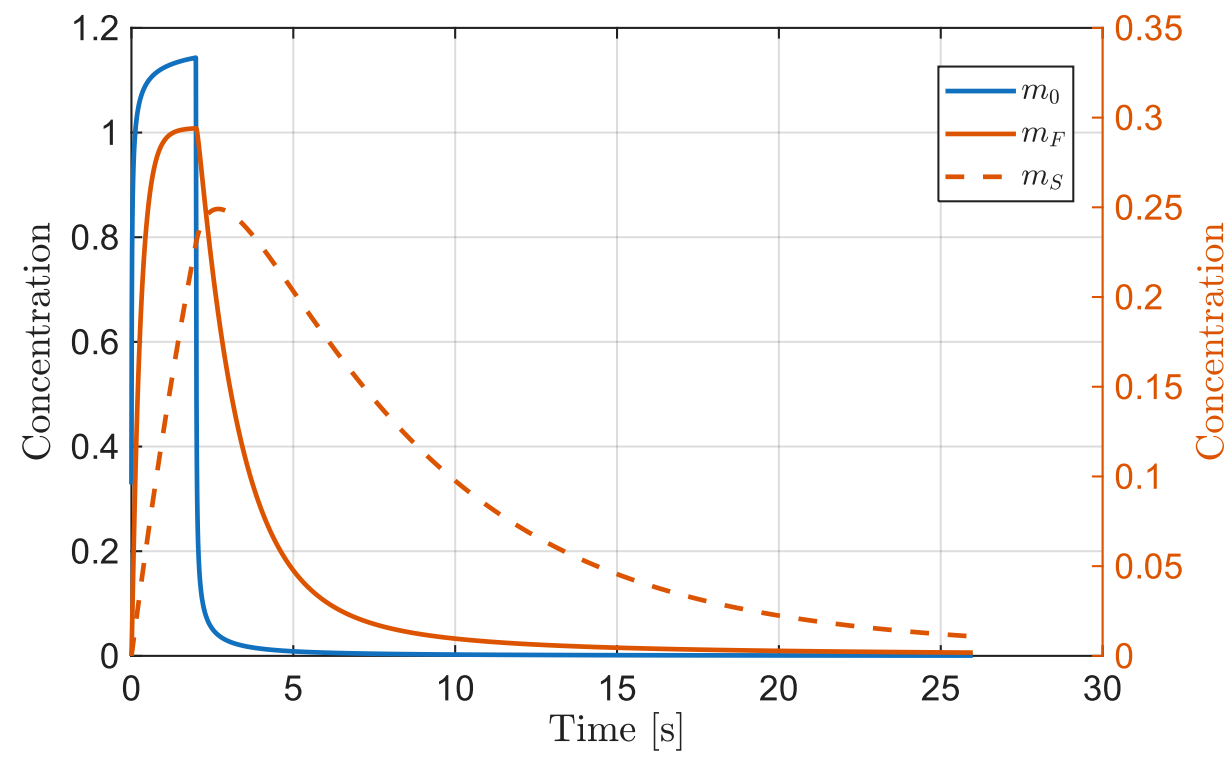
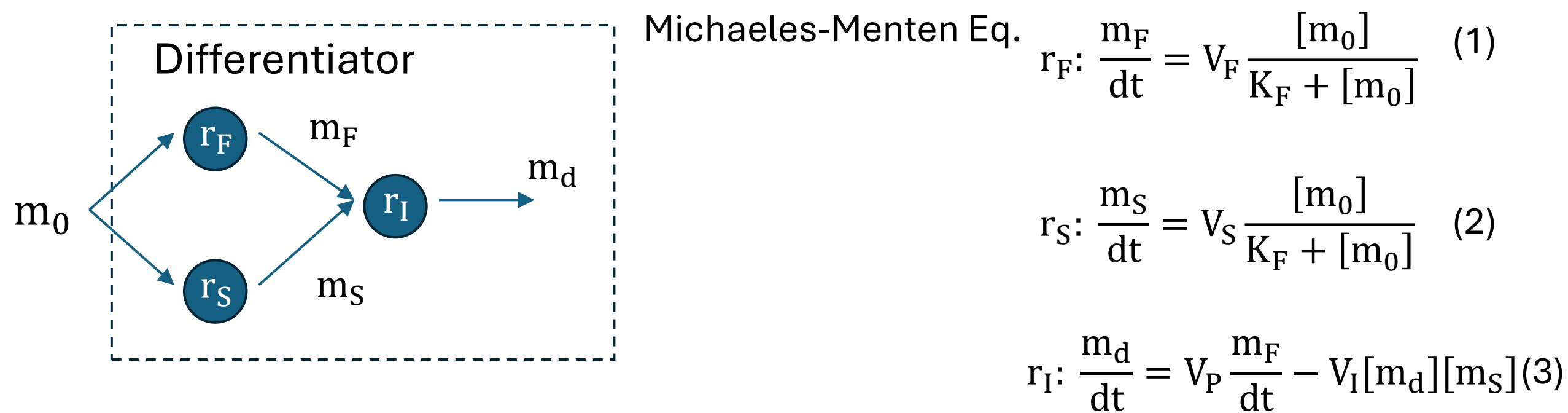




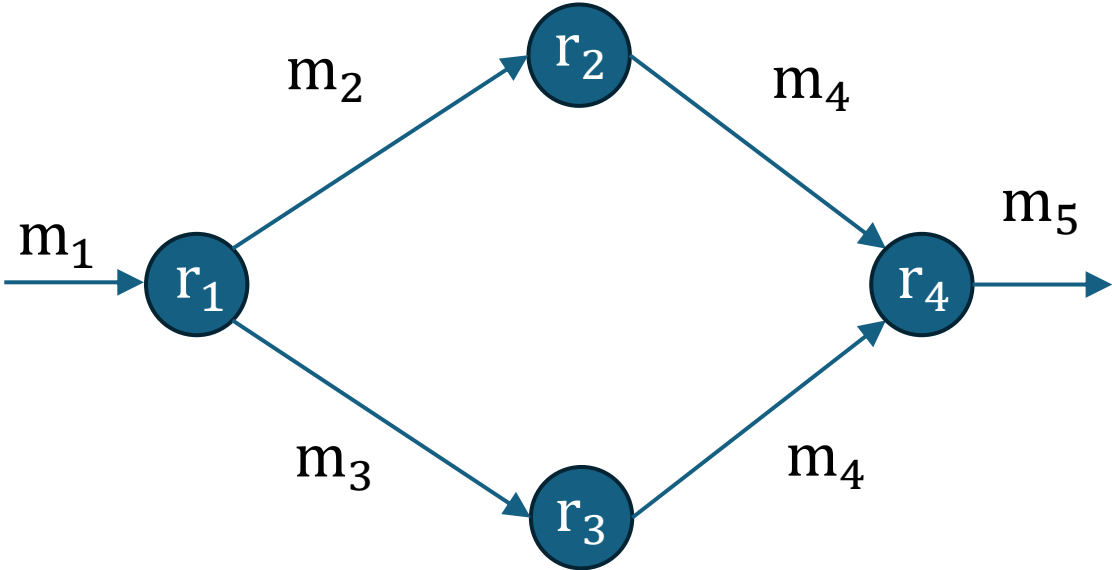
Point Transmitter

Receiver





Mapping a 2-neuron NN



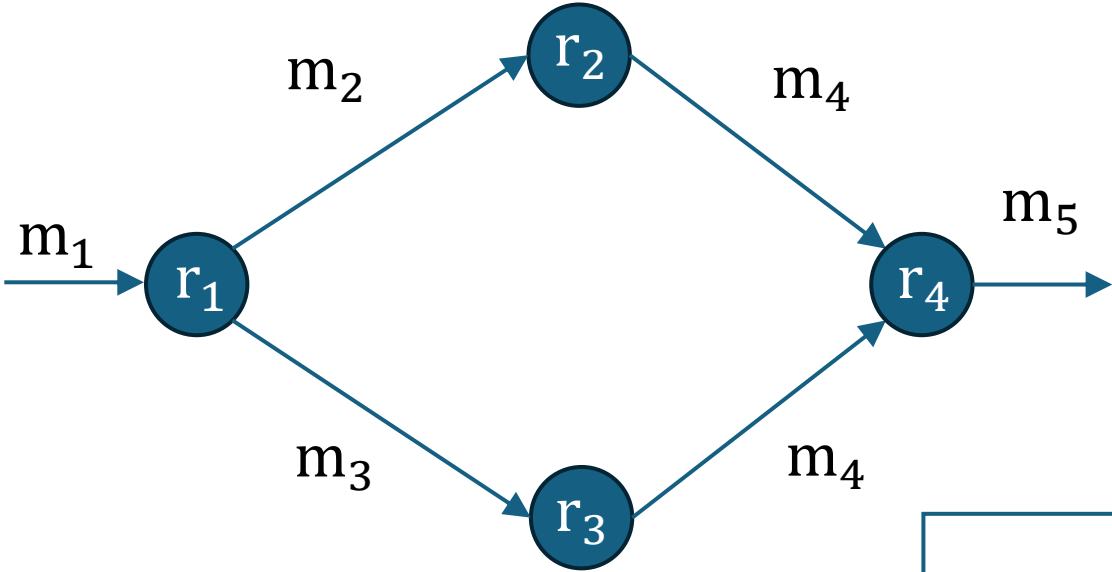
Pattern within the Stoichiometry Matrix

	r_1	r_2	r_3	r_4	r_5
m_1	-				
m_2	+	-			
m_3	+		-		
m_4		+	+	-	
m_5				+	

r_i r_j r_k

$[0, \cdots, b_{mi} > 0, \cdots, b_{mj} > 0, \cdots, b_{mk} < 0, \cdots, 0]$

Step 1: Find the m4-like metabolites and the reactions r₂'s and r₃'s

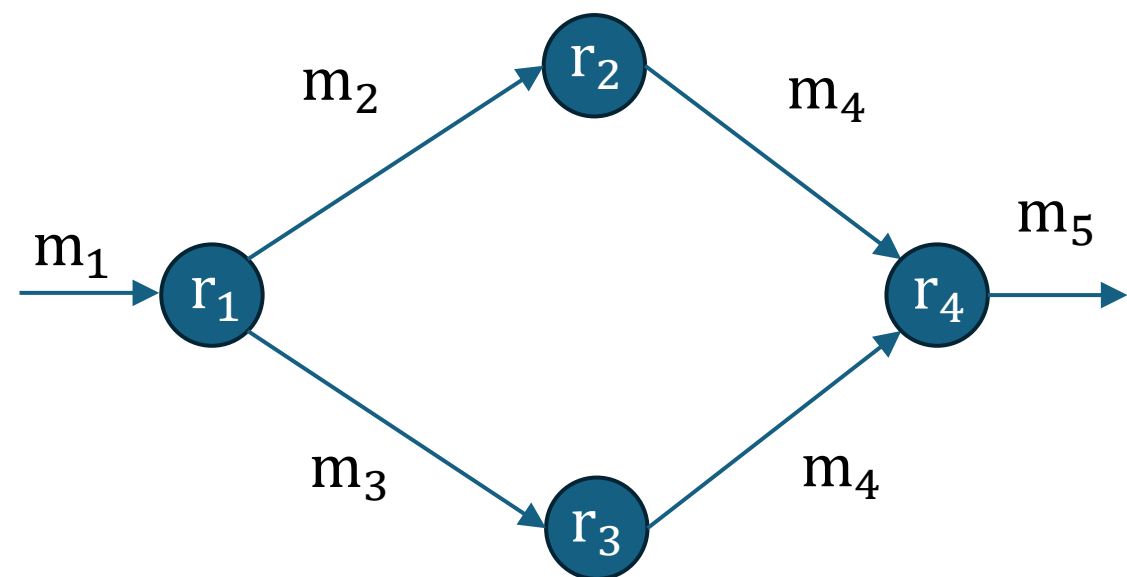


Pattern within the Stoichiometry Matrix

	r_1	r_2	r_3	r_4	r_5
m_1	-				
m_2	+	-			
m_3	+		-		
m_4		+	+	-	
m_5				+	

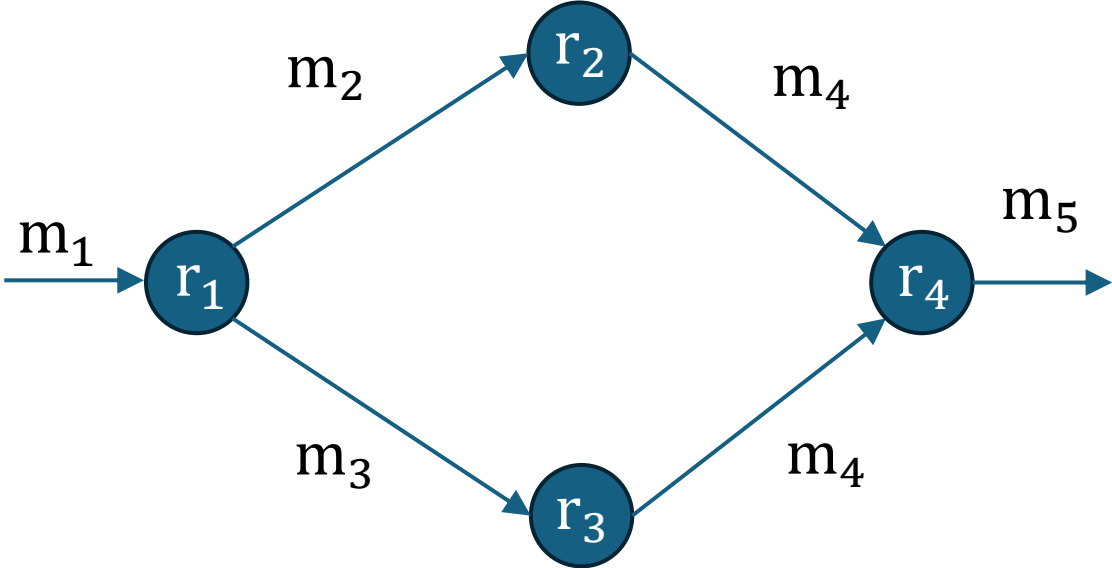
$$[b_{mi}, \dots, \boxed{b_{42} > 0}, \dots, \boxed{b_{43} > 0}, \dots, b_{44} < 0, \dots, b_{mR}]$$

Step 1: Find the m4-like metabolites and the reactions r_2 's and r_3 's



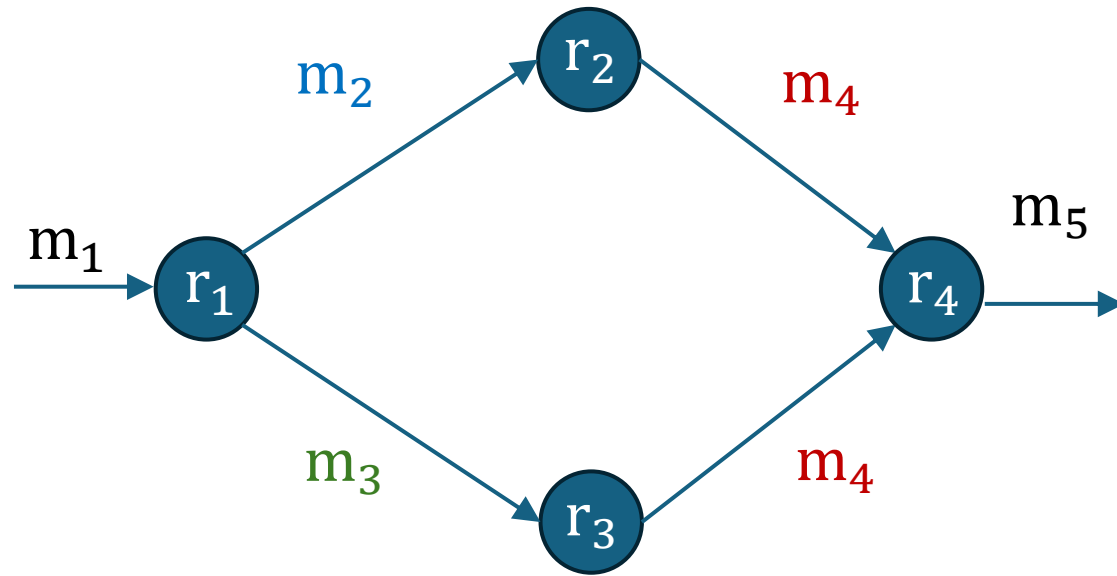
	m4	met	r2	r3	genes2	genes3	gpr2	gpr3
1	18	"s_0019"	591	592	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
2	18	"s_0019"	591	593	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
3	18	"s_0019"	591	594	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
4	18	"s_0019"	592	593	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
5	18	"s_0019"	592	594	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
6	18	"s_0019"	593	594	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
7	19	"s_0020"	3	78	"YAL060W"	["YGR087C"; "YLR...	"YAL060W"	"YGR087C or YL...
8	19	"s_0020"	3	731	"YAL060W"	["YGR087C"; "YLR...	"YAL060W"	"YGR087C or YL...

Step 2: Find reaction r_1 - like



	r_1	r_2	r_3	r_4	r_5
m_1	-				
m_2	+	-			
m_3	+		-		
m_4		+	+	-	
m_5				+	

Solutions



Metabolite m2: **C21H32N7O16P3S** | Name: coenzyme A| 405 | ID: s_0534

Metabolite m3: **C10H19O2** | Name: decanoate| 460 | ID: s_0600

Metabolite m4: **C31H50N7O17P3S**

Reaction 1: peroxisomal acyl-CoA thioesterase

EC code: 3.1.2.2

Gene: **YJR019C**

Formula:

$\text{C}_{31}\text{H}_{50}\text{N}_7\text{O}_{17}\text{P}_3\text{S} + \text{H}_2\text{O} \rightarrow \text{C}_{21}\text{H}_{32}\text{N}_7\text{O}_{16}\text{P}_3\text{S} + \text{C}_{10}\text{H}_{19}\text{O}_2 + \text{H}$

Location: Peroxisome

Reaction 2: acetyl-CoA C-acyltransferase (decanoyl-CoA)

EC code: 2.3.1.16

Gene: **YIL160C**

$\text{C}_{33}\text{H}_{52}\text{N}_7\text{O}_{18}\text{P}_3\text{S} + \text{C}_{21}\text{H}_{32}\text{N}_7\text{O}_{16}\text{P}_3\text{S} \rightarrow \text{C}_{23}\text{H}_{34}\text{N}_7\text{O}_{17}\text{P}_3\text{S} + \text{C}_{31}\text{H}_{50}\text{N}_7\text{O}_{17}\text{P}_3\text{S}$

Location: Peroxisome

Reaction 3: fatty-acid--CoA ligase (decanoate)

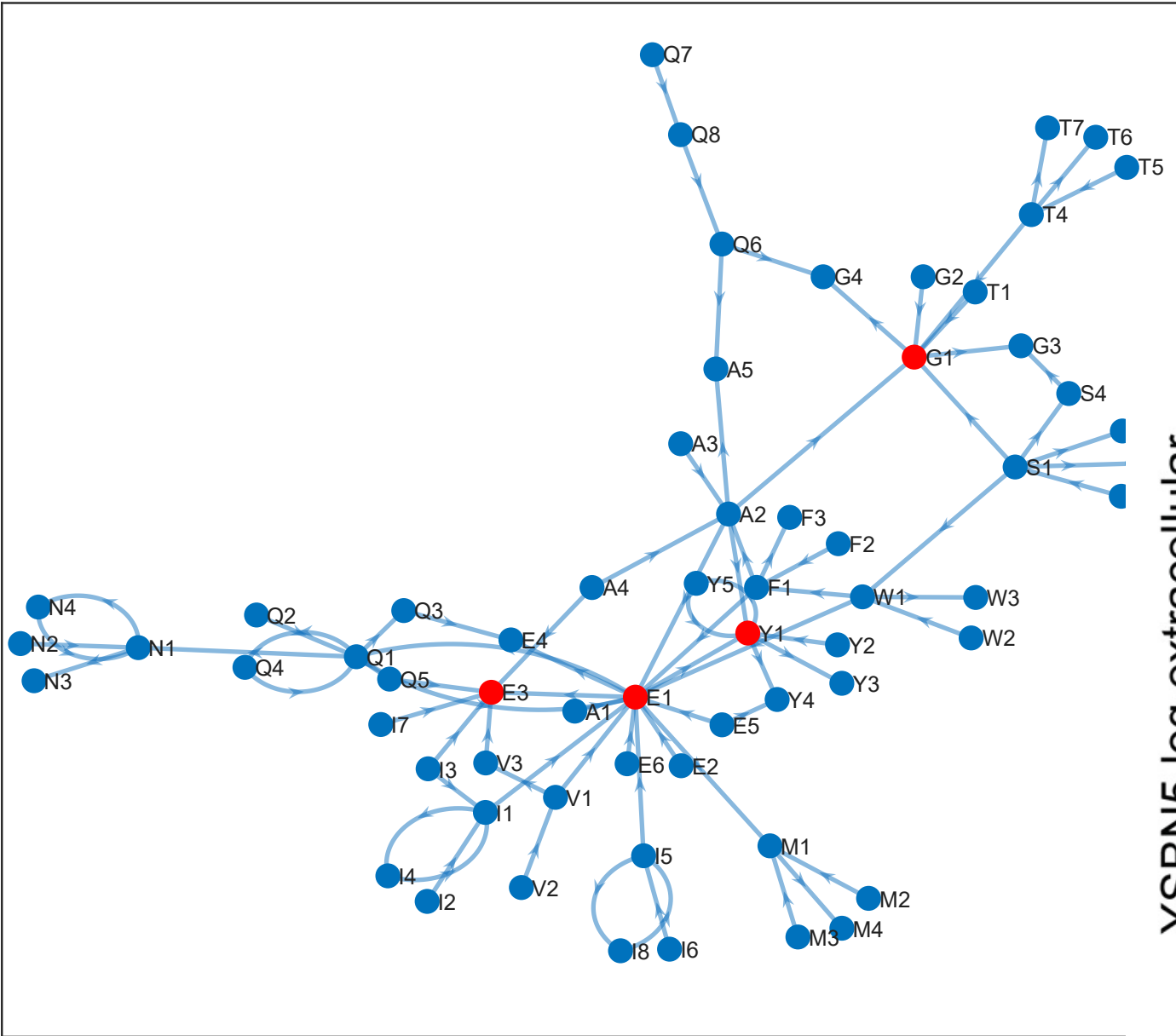
EC code: 6.2.1.3

Gene: **YER015W**

$\text{C}_{10}\text{H}_{12}\text{N}_5\text{O}_{13}\text{P}_3 + \text{C}_{21}\text{H}_{32}\text{N}_7\text{O}_{16}\text{P}_3\text{S} + \text{C}_{10}\text{H}_{19}\text{O}_2 \rightarrow \text{C}_{10}\text{H}_{12}\text{N}_5\text{O}_7\text{P} + \text{C}_{31}\text{H}_{50}\text{N}_7\text{O}_{17}\text{P}_3\text{S} + \text{HO}_7\text{P}_2$

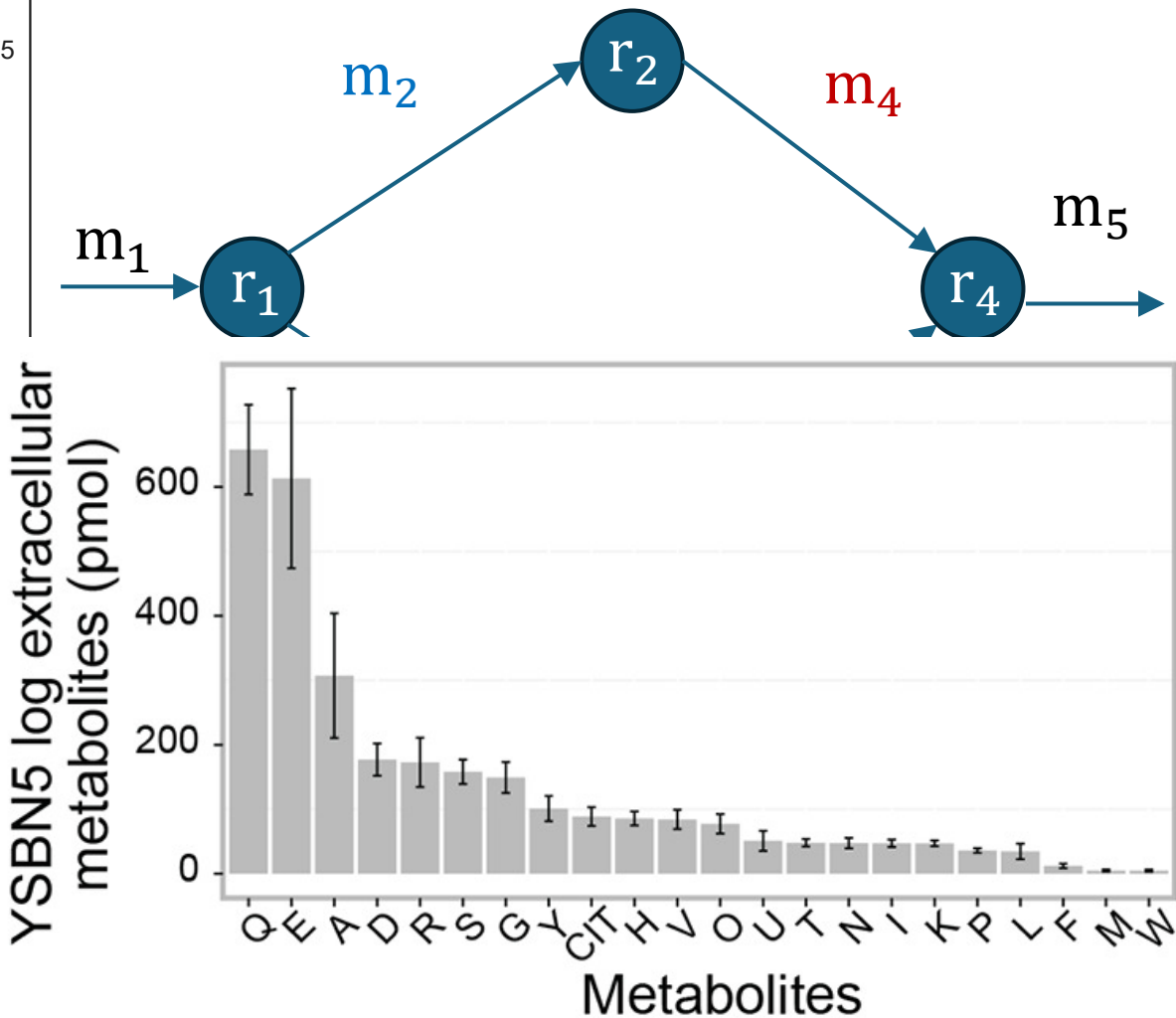
Location: Peroxisome

Mapping with restrictions

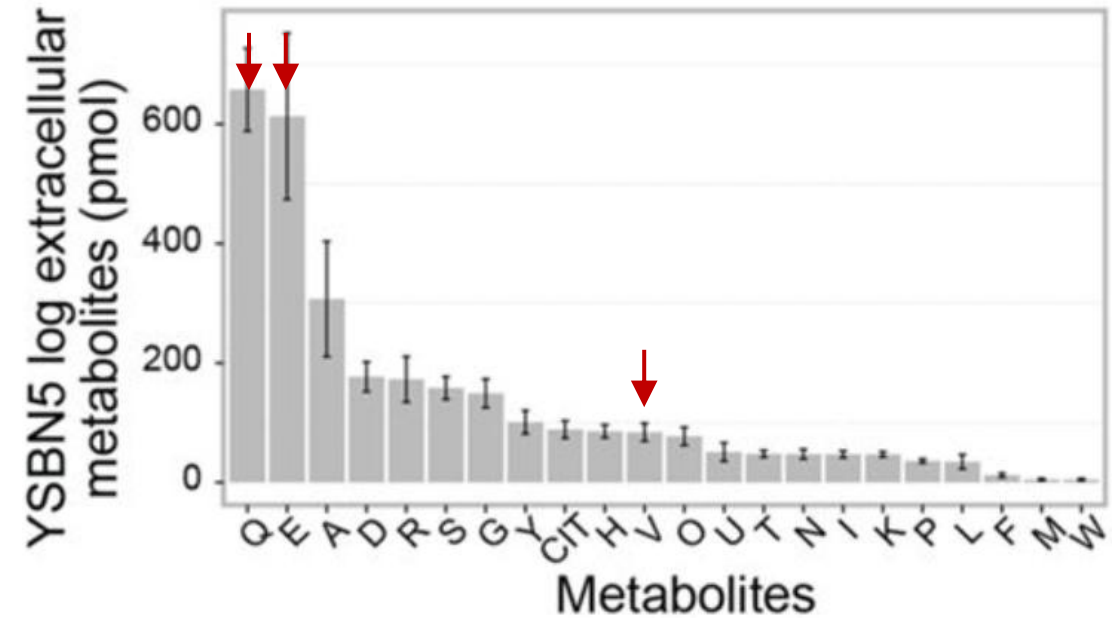


Restrictions:

1. The metabolite m_4 is a rich extracellular metabolite.
2. The metabolic network associated with m_4 is the least impactful.



1. The metabolite m_4 is a rich extracellular metabolite.
2. The metabolic network associated with m_4 is the least impactful.



Kate Campbell, Jakob Vowinckel, Michael Mülleder, Silke Malmshheimer, Nicola Lawrence, Enrica Calvani, Leonor Miller-Fleming, Mohammad T Alam, Stefan Christen, Markus A Keller, Markus Ralser (2015) Self-establishing communities enable cooperative metabolite exchange in a eukaryote *eLife* 4:e09943

THz-Induced Enzyme Unfolding Model

$$p_F = \frac{1}{1 + e^{\left(\frac{\Delta E - \langle E_{tot} \rangle_{ss}}{k_b T}\right)}} \quad (1)$$

$$\Delta E = E_f - E_u : \text{free energy} \quad (2)$$

$$\langle E_{tot} \rangle_{ss} = \frac{\alpha}{4\beta m [(\omega_o^2 - \omega_a^2)^2 + (\omega_a \bar{\gamma})^2]} (\omega_a^2 + \omega_o^2) + k_b T \quad (3)$$

$$\omega_o^2 = \frac{k}{m}, \omega_o \text{ is the natural frequency of the protein} \quad (4)$$

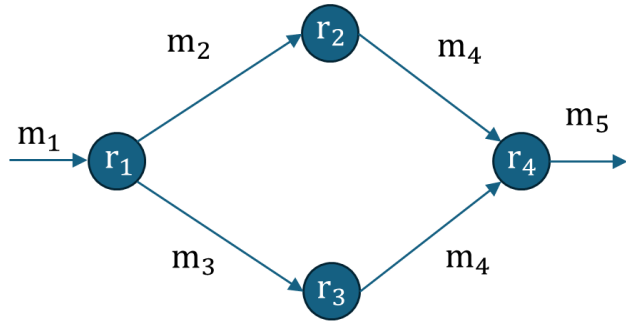
$$\bar{\gamma} = \frac{\ddot{\beta}}{m} \quad (5)$$

$$\ddot{\beta}: \text{Damping coefficient of the protein} \quad (6)$$

$$m: \text{Protein mass} \quad (7)$$

$$k: \text{Protein spring constant} \quad (8)$$

$$\omega_a: \text{Antenna radiation angular frequency} \quad (9)$$

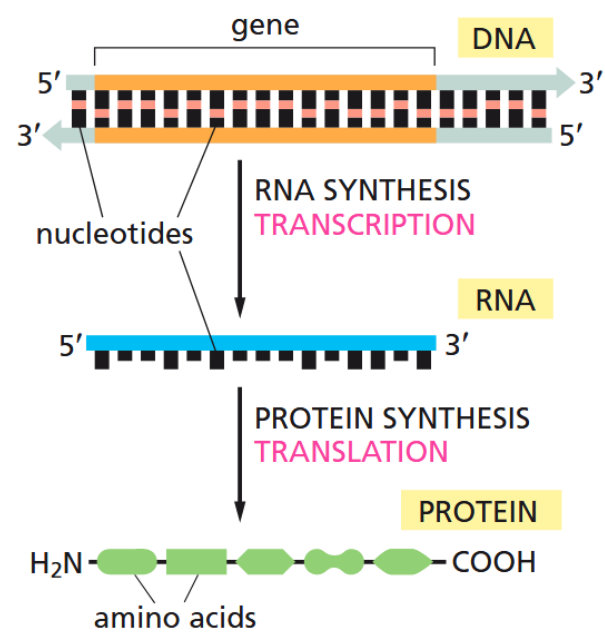


THI2

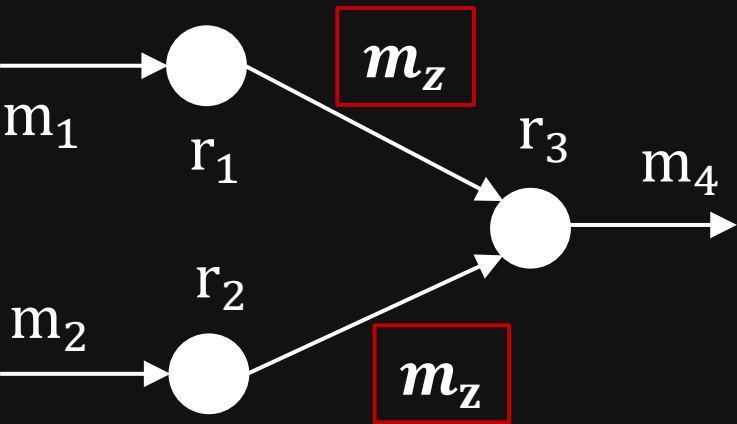
k: Protein spring constant

Physical Details	
Length (a.a):	450
Molecular Weight (Da):	51835.4
Isoelectric Point (pI):	8.63
Formula:	C ₂₂₈₅ H ₃₆₂₅ N ₆₃₅ O ₆₇₈ S ₃₁
Aliphatic Index:	76.96
Instability Index:	45.82

Looking for 2-neuron NNs



[1] B. Alberts, D. Bray, and K. Hopkin, Essential Cell Biology, 4th rev ed. edition. New York, NY: Norton & Company, 2014.



=== Motif 4 ===

Adder metabolite: s_{0293}

r1 (input): r_{0556}

$s_{0310} + s_{0434} \rightarrow s_{0293} + s_{0394} + s_{0794}$

$C_6H_9NOS + C_{10}H_{12}N_5O_{13}P_3 \rightarrow C_6H_8NO_4PS + C_{10}H_{12}N_5O_{10}P_2 + H$

gene: YPL214C

EC: 2.5.1.3;2.7.1.50

Location: cytoplasm

Subsystem: Thiamine metabolism

r2 (input): r_{4781}

$s_{0803} + s_{3910} \rightarrow s_{0293} + s_{0423} + s_{0456} + s_{0794}$

$H_2O + C_{17}H_{19}N_6O_{12}P_2S \rightarrow C_6H_8NO_4PS + C_{10}H_{12}N_5O_7P + CO_2 + H$

gene: YGR144W

EC: none annotated

Location: cytoplasm

Subsystem: Thiamine metabolism

z (output): r_{1036}

$s_{0267} + s_{0293} + s_{0794} \rightarrow s_{0633} + s_{1497}$

$C_6H_8N_3O_7P_2 + C_6H_8NO_4PS + H \rightarrow H_2O_7P_2 + C_{12}H_{16}N_4O_4PS$

gene: YPL214C

EC: 2.5.1.3;2.7.1.50

Location: cytoplasm

Subsystem: Thiamine metabolism